Stock Market Concentration and Portfolio Diversification

ABSTRACT

This paper investigates the underlying effects of stock market concentration on the portfolio diversification potential. Using four decades of data from the United States on quarterly frequency, we reveal that the concentration in stock market dominated by a small group of large and successful companies is negatively correlated with portfolio diversification potential both in the long run and short run with strong statistical significance. This negative effect is notably larger in the longer term. Our findings are robust to alternative measurement and market portfolio. Besides, we also test for the components of portfolio diversification potential separately, uncovering the significant relations between stock market concentration and these components. The time variations of market-level and firm-level volatility series are also updated in this paper for the recent twenty years, and small, negative but insignificant trends are captured during this time period.

Keywords: Stock market concentration, Portfolio diversification, Stock market volatility, Idiosyncratic risk

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Statement of Originality

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I declare that the text and the work presented in this document are original and that no sources other than those mentioned in the text and its references have been used in creating it.

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1 Introduction

The COVID-19 pandemic has caused dramatic effects on the financial markets. The contractions in economic and social activities have significantly undermined economic growth and caused unprecedented shocks to the financial market globally (Baker et al., 2020; Fernandes, 2020). The unknown risks and challenges are not only posed for the entire market and industries, but also break the original expectations of the investors. Aside from this novel infectious disease outbreak, a large and expanding literature points out that historical recessions also led to massive instability in the economy and stock markets (Caballero & Hammour, 1991; Gómez-Cram, 2022). The extreme uncertainties in the risk and returns of stocks have forced the investors to reallocate their portfolios (Himanshu et al., 2021). Thus, in this environment of ongoing uncertainties, demand for broadly diversified portfolios has reached an unprecedented level in an effort to minimize the associated risks.

In addition to the overall impact of the business cycle, the market characteristics also significantly impact portfolios' construction and diversification. Under the dynamic economy nowadays, it can be obviously noticed that businesses are continuously urged to compete for consumers since they are faced with the constant threat of new incomers entering the markets. To be the winners or even just survivors under the market competition, firms are forced to constantly keep innovative and upgrade the products' qualities while offer more competitive prices. Otherwise, they will be easily replaced by those who are able to. Naturally, some firms that are more innovative and productive stand out in the markets. And this will be gradually followed by the phenomenon of market concentration if a group of very successful companies earn higher returns than other competitors in the markets. In this study, we shed light on the stock market concentration. The stock market is critical to the functioning of economy since it distributes capital to various parties, provides necessary funding to businesses and so on. Hence, the occurrence of concentration inevitably results in a substantial impact on the stock market as a whole and also on individual stocks. Given the phenomenon that markets are getting concentrated and the increasing demand for portfolio diversification nowadays, it is rather worthwhile to make sense of the relationship between these two phenomena. Therefore, this paper aims to address the extent to which the stock market concentration can affect the degree of portfolio diversification. Specifically, the following research question will be tested:

"How does the increased stock market concentration affect the portfolio diversification potential?"

Inspired by the study of Bae et al. (2020), the regression model of ordinary least squares is adopted with stock data from United States for a 45-year period. As mentioned above, we look at the stock market concentration, that is, how much of the total stock market value is made up of a small number of very large companies. The degree of portfolio diversification is considered as the inverse of the normalized portfolio variance of value-weighted market portfolio, measured as the ratio of average variance of individual stocks over the market portfolio variance. According to this measurement, how much the portfolio variance changes can be captured by adding one more security. Moreover, the effects of some macroeconomic factors are also controlled in the regressions. Based on the regression results under different time lags, we find out that the stock market concentration is significantly negatively related with portfolio diversification potential in both long and short runs. And the magnitude of this negative effect keeps eliminating in absolute terms until the contemporary year. Further, we run additional tests to explore the underlying reasons behind this significantly negative relation. The outputs show that stock market concentration is significantly related to the sub-components in portfolio diversification potential, especially the numerator average variance of individual stocks. Except the component testing, we perform both graphic and trend analyses capturing the time trends of these components of diversification potential, updating the results of Campbell et al. (2001). Last but not least, two additional robustness checks are implemented by adopting different measurement and market portfolio. Both tests confirm our findings from the main analysis.

Understanding the link between stock market concentration and portfolio diversification provides implications for individual investors and portfolio managers to seek sufficient and even maximum diversification under the current environment with more concentration but higher diversification demand. To date, there remains a surprising paucity of empirical research directly linking stock market concentration and portfolio diversification. Most studies in the field of portfolio diversification have focused in a setting of international portfolio, mostly looking at both global and local factors such as the developments and integration of the overall international market (Eiling et al., 2012; Lessard, 1973). Although there are much research focusing on the market concentration, most are undertaken for the industry analysis, investigating the level of concentration in different industrial markets (Grullon et al., 2019). Up to now, stock market concentration has attracted some discussions but generally in a macroeconomic context, such as its relation with the real economy (Bae et al., 2020). In terms of the stock market volatility, the testing period of previous studies are largely restricted to the 20th century. Thus, this paper is aimed to fill a gap in the literature by examining the unclear relation between stock market concentration and portfolio diversification potential. Additionally, this work will also present fresh insights into the stock market volatility and the average volatility of individual stocks for a more recent time period.

The paper proceeds as follows. The overall structure of the study takes the form of six main sections. In Section 2, the prior literature is explicated first to offer a fundamental basis for research contributions. The third section lists the sample and databases used along with reports and analyses of summary statistics. Additionally provided is a thorough explanation of the research methodology, including both variable constructions and regression models. Next, section 4 analyzes the results obtained and addresses the research question, followed by the section regarding robustness checks. Lastly, the final section summarizes the principal findings of this research, discusses the corresponding constraints while giving implications for future studies.

2 Literature Review

2.1 Factors affecting portfolio diversification

To provide the empirical framework for main analysis, we first embark on the existing literature discussing the underlying factors impacting portfolio diversification level. Growing literature advocates that the degree of portfolio diversification lies in investors' own characteristics. For instance, some investors have affinities to some specific security characteristics and some certain behavioral biases such as overconfidence and trend-following behavior (Goetzmann & Kumar, 2008). Moreover, with socioeconomic and behavioral differences controlled, Abreu & Mendes (2010) documented that the level of investors' financial literacy has positive impacts on the degree they choose to diversify the portfolios. Some literature also emphasizes the vital role of portfolio size in addition to investor characteristics. A relatively stable and predictable correlation has been found between the number of securities in the portfolio and the portfolio dispersion degree (Abreu & Mendes, 2010). Due to the fact of the increase in individual stock volatility over the last few decades in the 20^{th} century, more stocks have been required to achieve any given level of portfolio diversification, and this diversification process proceeds rather rapidly as portfolio size grows (Campbell et al., 2001; Klemkosky & Martin, 1975). Apart from that, the role of stock correlations is also noteworthy. The decreasing correlations between stocks are related to the increasing benefits of portfolio diversification (Campbell et al., 2001). By comparing the risk exposures of investors' portfolios and those of benchmark portfolios, Goetzmann & Kumar (2008) suggested that, to a large extent, the changes in diversification characters of investors' portfolios are affected by the changes in the correlation structure of the US equity market.

2.2 Aggregate stock market volatility

Because portfolio diversification aims to minimise underlying risk exposures of the portfolio, this study also pertains to the literature on the components of stock volatility. Aggregate market volatility refers to the volatility suffered by investors of aggregate index funds and plays an essential role in almost any financial theory of risk and return. Countless papers also stress the crucial relationship between stock market volatility and stock returns. French et al. (1987) pointed out the positive relation between the volatility of a stock market portfolio and portfolio's expected risk premium. Likewise, Pindyck (1984) stressed that the transition to highly unstable periods can explain the very poor performance of the stock market, especially in the 1970s.

Moving to the specific statistics and trends, the market volatility was particularly high around the 1970s and 1980s, as shown in the work of Campbell et al. (2001), spiking in October 1987, during which the stock market crash took place. Also confirmed by Officer (1973), the variability of stock return was unusually high during the Great Depression from 1929 to 1939. Hence, stock market displays a pronounced business cycle pattern, being higher during recessions and lower for expansions (Brandt & Kang, 2004; Hamilton & Lin, 1996). This phenomenon is also reflected in the perspectives from Schwert (1989), stating that stock market volatility is relevant to the general health of the economy. Meanwhile, a large and growing literature has demonstrated that the economy-wide factors are responsible for the movements in stock market volatility. In a study conducted by Engle et al. (2013), it is shown that macroeconomic fundamentals play a significant role even in the short run. On a daily scale, roughly 10% to 35% of expected one-day-ahead volatility is accounted for elements such as inflation and industrial production growth. In addition to macroeconomic variables, the time variation of stock market volatility can also be attributed to the stock trading activities. For instance, large price movements tend to cause large trading volume. Evidence presented in the existing literature favors the proposition that stock market volatility appears higher when trading volume increases (Schwert, 1989).

2.3 Idiosyncratic risks in the individual stocks

Nevertheless, the aggregate market return does not account for the total return of an individual stock. Campbell et al. (2001) revealed that the average stock returns of individual stocks were becoming more volatile over the time from 1962 to 1997. Specifically, it is the firm-level volatility which trended upward during that period while the volatilities of both market and industry remained relatively stable. Similar to how stock market volatility increases significantly, the considerable surge in firm-level volatility also occurs during economic downturns that eventually lead to recessions. Up to a one-year lead and lag, the firm-level volatility embodies a negative correlation with GDP growth, which is also presented in the cyclical behavior of both market and industry volatility series. To explain the growing volatility on firm level, research also found out that newly listed firms contribute more to the upward trend (Wei & Zhang, 2006). Their profitability is mostly declining, but the growth potentials remain strong. Besides, average return on equity is also useful for explaining the positive trend in average variance of individual stocks (Wei & Zhang, 2006). All these findings presented offer profound implications for investors since the prior study also pointed out that the increasing idiosyncratic risks of individual stocks significantly corrode the risk-reward relationship (Campbell et al., 2001).

2.4 Changes in markets - the growth of large companies

To further identify the changes occurring in the market, we first shed light on the rapid growth of some large and successful firms. Growing literature highlights the fundamental roles of large companies in the economy. They account for a substantial portion of a country's imports, exports, and dominance in key areas such as investments, research and so on. In 2016, businesses in US spent around $375 billion solely on R&D performance. And specifically, those large firms with more than 5,000 employees contribute almost two thirds of that huge amount (Wolfe, 2018). Notable is the fact that some idiosyncratic shocks to these huge enterprises can even induce fluctuations in a nation's GDP (Gabaix, 2011).

Contrary to the commonly held belief that these top and large enterprises are experiencing enormous productivity gains and play a more essential role in the economy, relevant literature finds contradictory evidence. In the work of Gutiérrez & Philippon (2019), with both firm-level and industry-level data, they found out that these companies did play a key role in promoting economic growth in the past. However, in recent years they are not contributing as much as they used to. In particular, their contribution to US productivity growth has fallen by more than a third since 2000. Similarly, Schlingemann & Stulz (2021) investigated the relation between listed firms in the stock market and the real economy, showing that compared with the 1970s, the listed firms contribute less to the real economy in terms of employment and GDP. Behind these contrary implications, the decline in the manufacturing industry and instead the rapid growth in the service industry can partially explain the current weaker and indirect link between listed firms and the real economy.

2.5 Changes in market – markets are becoming concentrated

Further, a small amount of large and successful firms which generally earn higher returns than other listed firms tend to inevitably bring out the market concentration. Fast-growing literature underscores that the US product markets are undergoing new fundamental changes. Research by Grullon et al. (2019) shows that the systematic increase in Herfindahl–Hirschman Index (HHI), which is the measure of market concentration, has appeared in more than 75% of the US industries over the last two decades, and the average growth in the market concentration level has markedly reached 90%. Next to the rapid growth in the concentration level, they also investigate the link between the firms in the industries and the corresponding concentration level, revealing that the firms in the industries with the largest concentration increase display the higher profit margin as well as more profitable M&A deals. However, it is also noteworthy to raise one striking phenomenon that United States now has abnormally few publicly listed companies. After the peak year in 1996, the new list rate in US stock market is gradually low while the delist rate shows a totally opposite tendency, which are both not witnessed in other countries (Doidge et al., 2017). Based on the available research, the recent listing gap that occurred in United States can be attributed to the low benefits from a US listing and simultaneously the unusual high rate of acquisitions among the publicly listed firms (Chaplinsky & Ramchand, 2012; Doidge et al., 2017).

2.6 Changes in markets – stock market concentration and its effects

In terms of the effects from stock market concentration, Bae et al. (2020) used the three-decade data from 47 countries and tested the stock market concentration on fundamental indicators of economic growth. Their theoretical implications stress the economically large negative effects of market concentration on the economic growth. To be specific, a concentrated stock market makes it difficult for new and innovative companies to enter the stock markets and get necessarily financed. Less efficient capital and slower economic growth are all the consequences related to the concentration in stock market. Plus, the negative effects can also be captured in the fewer IPOs and innovation activity in these concentrated markets. For instance, those highly successful companies can deploy their economic and political power to achieve a constant advantage via suppressing new competitors and their access to capital funding (Braun & Raddatz, 2008). Agarwal et al. (2014) also mentioned that the stock exchanges dominated by several large and successful corporates tend to become more sensitive and even inclined towards these market leaders. Because any of them can easily affect the profitability of these stock exchanges. Under the dominance of these leading companies, the needs of some start-ups to raise equity will be addressed with less sensitivity in the stock exchanges.

In summary, empirical literature mentioned gives clear implications that markets are turning more concentrated due to the dominance of a small group of firms with large market capitalization, strong competitiveness, and successful operations. The phenomenon of market concentration exists across various industries. With respect to the stock market, the concentration in this essential player in the economy strikingly presents its negative influence on the indicators in real economy. In other words, stock market concentration partially contributes to the poor performance of the economy.

Besides that, start-ups and firms with a low market capitalization in the stock market also suffer from impacts of stock market concentration. Meanwhile, a large and growing literature highlights the underlying influence factors on portfolio diversification potential. From the individual perspective, portfolio size and investor characteristics such as the level of financial literacy function critically when diversifying the portfolios. The correlation structure of the equity market, business cycle factors and trading activities all connect to the portfolio diversification potential in the macroeconomic context. Following the stock market concentration's negative effects on economic-wide elements, we can come up with the conjecture that stock market concentration may also underlyingly have adverse impacts on the portfolio diversification potential. Because diversification potential is associated with certain macroeconomic factors. To address the research question, the following hypothesis will be tested for both main analysis and robustness checks:

"The increased stock market concentration is negatively correlated with the portfolio diversification potential."

3 Data & Methodology

In this section, the information of sample and the corresponding data resource are firstly described, followed by the selection of the market index and database for control variables. Following that, a comprehensive outline of the methodology utilized in this paper is next provided. It commences with the variable constructions, and then models for both trend analysis and main regressions will be introduced. Lastly, the summary statistics are also addressed, covering all the variables under examination and relevant NBER recession dates.

3.1 Sample Data

Data for this research are collected from the Center for Research in Stock Prices (CRSP) at the University of Chicago. We use the daily return data of firms traded on the NYSE, the AMEX and the Nasdaq to compute the quarterly variables. Considering the data availability of control variables, the sample runs from 1975 to 2019. Meanwhile, the information regarding daily prices and the number of shares outstanding for the same time period are also obtained in the CRSP data set. During this 45-year period, the maximum number of firms contained in the sample was 8090 in 1996, while in 1979 there were only 2041 listed companies included. For the market portfolio, we retrieve daily data from NYSE/AMEX/ Nasdaq composite index, which is also provided in CRSP dataset. In line with the paper from Campbell et al. (2001), value-weighted returns are drawn from this market index for the period from 1975 to 2019. With respect to the control variables, data on the proportions of the total value of traded stocks in US stock market on the GDP is drawn from the World Development Indicators[1] provided by the World Bank. Data about the rest of control variables for the same period is all retrieved from Refinitiv Eikon. In particular, due to the data availability of inflation rate on a quarterly level, the inflation rate is chosen to be measured by the quarterly GDP price deflator in this study.

3.2 Variable Constructions

3.2.1 Stock Market Concentration

The measurement for the degree of stock market concentration is prepared according to the procedures used by Bae et al. (2020). First of all, the market capitalization for each firm is computed as follows with daily stock data:

[1] For download of detailed data, see: https://databank.worldbank.org/source/world-development-indicators#

$$MC\_Firm_{i,d} = Price_{i,t} \times Share\_Outstanding_{i,t}$$

where i denotes the firm i and d means day d. After calculating the daily market capitalization, we extract the market capitalization values for each company when the corresponding months are displayed as March, June, September, and December. Furthermore, only the end-date data for these months will be retained.

According to that, we can next obtain the sum of the largest five firms' market capitalizations on a quarterly frequency with this equation:

$$MC\_Top5_{q,t} = \sum_{i=1}^{5} MC\_Firm_{i,q,t}$$

In this formula, t means the year t, and q represents the quarter q in year t. To be specific, q equals 1 when the month is March, 2 when it is in June, 3 when September is shown in month, and 4 for December data.

Similarly, the aggregate stock market capitalization can be simply gained by summing up the market capitalization of all the individual firms.

$$MC\_Total_{q,t} = \sum_{i=1}^{n} MC\_Firm_{i,q,t}$$

where n is the number of firms contained in the domestic stock market for quarter q in year t.

To arrive the stock market concentration, the formula from the paper of Bae et al. (2020) is utilized below, indicating the proportion the largest firms account for in the domestic stock market. In this case, we choose the top 5 firms in the rank.

$$Mkt\_Con\_Top5_{q,t} = \frac{MC\_Top5_{q,t}}{MC\_Total_{q,t}}$$

Following the existing literature (Bae et al., 2020), we also take into account another choice of taking the sum of the largest ten firms' market capitalizations as the numerator in the formula above. Similar to the variable with the top five corporates, the measurement of market concentration can be elaborated as follows:

$$Mkt\_Con\_Top10_{q,t} = \frac{MC\_Top10_{q,t}}{MC\_Total_{q,t}}$$

3.2.2 Portfolio Diversification Potential

To accurately capture the extent of diversification potential in portfolios, we adopt the method in the work of Goetzmann & Kumar (2008). The normalized portfolio variance (NV) is firstly generated by dividing the portfolio variance by the average variance of individual stock contained in that portfolio. Intuitively, the measure of normalized portfolio variance (NV) reflects the level of reduction in portfolio variance by increasing the number of stocks included in the portfolios. As confirmed by the prior research, the normalized variance decreases as the number of stocks in stocks in the portfolio rises. A better-diversified portfolio is expected to have a lower normalized variance than a more concentrated portfolio (Goetzmann & Kumar, 2008). Hence, for easier understanding and more direct interpretation, we take the inverse of the normalized portfolio variance as the natural method measuring the portfolio diversification potential. The corresponding formula is displayed as follows:

$$Div_{q,t} = \frac{1}{NV_{q,t}} = \frac{ave\_var_{q,t}}{mkt\_var_{q,t}} = \frac{\overline{\sigma^2}_{q,t}}{\sigma^2_{port\,q,t}}$$

where q and t represent the quarter q and year t, respectively. In this case, Div in the formula is the direct measurement of the level of portfolio diversification potential. Higher Div represents higher portfolio diversification potential directly and simultaneously lower normalized portfolio variance. In our study, we also aim to measure the portfolio diversification potential in a more general way. In light of this consideration, we set the portfolio as the market portfolio, and the individual stocks in it are basically all the stocks of which the stock market is comprised. In line with the paper from Campbell et al. (2001), we obtain value-weighted returns of the market portfolio to calculate the market portfolio variance:

$$\sigma^2_{port\_value_{q,t}} = \frac{\sum_{h=1}^{h}(R_{value_{d,q,t}} - \overline{R_{value}}_{d,q,t})^2}{h}$$

where $\sigma^2_{port\_value_{q,t}}$ is the variance for value-weighted market portfolio in quarter q of year t. We take the quarterly variance over the daily returns contained in the corresponding quarter. Namely, all three months' daily returns are taken into consideration for each quarter. Hence, $R_{value_{d,q,t}}$ represent the returns on day d in quarter q of year t, including all distributions, on the value-weighted market portfolios. Furthermore, the symbol h means the number of daily returns included in the quarter q of year t.

In terms of the average stock volatility, it is measured by the mean of variances of all individual stocks included in the stock market:

$$\overline{\sigma^2}_{q,t} = \frac{1}{n}\sum_{i=1}^{n}\frac{\sum_{h=1}^{h}(R_{i,d,q,t} - \overline{R_{i,d,q,t}})^2}{h}$$

where $R_{i,d,q,t}$ denotes the daily return for stock i in quarter q of year t. In other words, the quarterly variance of each stock in year t is firstly computed with its available daily stock returns, and then we take the average of variances across all the individual stocks in an equally weighted way (Goetzmann & Kumar, 2008).

3.3 Regression Models

3.3.1 Trend Analysis

One of the purposes of this paper is to update the work of Campbell et al. (2001) with fresh insights into stock market volatility and average volatility of individual stocks for a more recent time period. Therefore, in this paper, we also plot and investigate the time-series trends of these two volatility series, which are also the variance components in the variable Div, supplementing the study of portfolio diversification potential. We firstly follow the paper from Bai & Perron (2003) and test for the date when structural breaks in volatility series take place. Next, we employ the trend regressions conducted in Campbell et al. (2001). Following the procedure suggested by Vogelsang (1998), trend regressions are carried out as follows:

$$mkt\_var_t = \mu + gt + \rho mkt\_var_{t-1} + u_t$$
$$ave\_var_t = \mu + gt + \rho ave\_var_{t-1} + u_t$$

where mkt_var_t and ave_var_t are the volatility series under examination, representing the market-level and firm-level volatility, respectively. μ means the intercept of this regression and g is the trend coefficient which is the key indicator of the trends. Besides that, ρ reveals the dependences of market-level volatility and firm-level volatility on their own first lags.

3.3.2 Main Regressions

In order to examine the association between increased stock market concentration and the level of portfolio diversification potential, the following regressions equation will be adopted and tested:

$$Div_{q,t} = \beta_0 + \beta_1 Mkt\_Con\_Top5(10)_{q,t-j} + \gamma_1 Inflation_{q,t} + \gamma_2 GDP\_Growth_{q,t} + \gamma_3 Listed\_Firms_{q,t} + \gamma_4 Stocks\_Traded_{q,t} + \epsilon$$

In this regression, the dependent variable $Div_{q,t}$ is the degree of portfolio diversification potential obtained in quarter q of year t based on the value-weighted market portfolio. In line with the literature (Bae et al., 2020), the independent variable

$Mkt\_Con\_Top5(10)_{t-j}$ means the lagged value of stock market concentration level in year $t-j$ with choices of both top 5 and top 10 firms tested. And j equals to the number of years lagged, ranging from 0 to 5. Using the lagged values enables us to better manifest and compare the effect of increased stock market concentration for both the long and short haul (Bae et al., 2020).

Aside from the explanatory variables, we also control for the following factors affecting the stock volatility. As indicated in the literature review, macroeconomic variables such as inflation account for partial variations in stock market volatility (Engle et al., 2013). Returns of common stocks also present a negative relation with the expected inflation rate. (Fama & Schwert, 1977) Therefore, $Inflation_{q,t}$, the quarterly inflation rate of United States as measured by the GDP price deflator, is included in the regression. Similarly, previous studies also demonstrate the importance of business cycle factors on stock market volatility. Even though the NBER dates can serve as a benchmark, the binary NBER classification scheme is still likely to miss some useful information (Campbell et al., 2001). Thus, we choose to include variable $GDP\_Growth_{q,t}$ to control the cyclical behavior of volatility, referring to the growth rate of US gross domestic product for quarter q in year t.

$Listed\_Firms_{q,t}$ represents the number of listed firms in the US stock market in quarter q of year t. Mentioned in the second section, Campbell et al. (2001) advocated that as idiosyncratic risk has grown over time, more stocks are required to achieve relatively complete portfolio diversification. In this case, the selection of this control variable captures the size of market portfolio and is supposed to affect the diversification. Moreover, the time variation of stock market volatility is also related to stock market trading (Schwert, 1989). To control this, we add another control variable $Stocks\_Traded_{q,t}$, measured as the proportion of the total value of traded stocks in US stock market on the GDP for each quarter in the corresponding year.

Additionally, the β_0 in the model means the constant and β_1 represents the coefficient between the dependent variable and the variable of interest to the analysis. Moreover, five γ indicate the coefficients for respective control variables. In the end, ϵ represents the error term for the regression. To be specific, if obtained β_1 shows a negative sign, then this result can be utilised to determine that the hypothesis should not be rejected. Simultaneously, the statistical significance also matters for presenting the underlying relationship. If no significance is shown in the results, the statistical influence of the obtained coefficients is cast in doubt.

3.4 Summary Statistics

3.4.1 Summary Statistics for Variables

Table 1 below presents the averages, maximums, and minimums of all the variables included in the regression model. Firstly, the summary statistics of normalized portfolio variance and portfolio diversification potential are reported in the first and second rows. It is evident from the standard deviation of 33.9884 that the degree of portfolio diversification potential varies greatly over time. And Figure 1, which charts the trend of portfolio diversification potential, demonstrates the same outcome. From this graph, it is apparent that there are a lot of fluctuations in the diversification level across the years. However, except for the spike in the 1990s, most are still fluctuating around the average value which reaches the high level at 32.6698, reflecting the perfect diversification of a market portfolio. Turning to the extent of stock market concentration in United States, more robust results can be obtained by including variables $MC\_Top5(10)$ with both the largest five and the largest ten corporates. The time-series trend of these two variables presented in Figure 2 reveals the general tendency of decreasing stock market concentration in US as time passes from 1975 to

2019. Combined with the statistics in Table 1, the US stock market was highly concentrated in the 4th quarter of 1975, peaking at 0.2032 (0.2749). Afterwards, the concentration of the US stock market declined and reached its lowest point of 0.0636 (0.1054), given the dramatic growth in the number of listed firms in the 1990s shown by Figure 5 from the Appendix and the statistics from Doidge et al. (2017). Then, the graph presents the growth of US stock market concentration from the end of the 20th century, but this steady increase ceased around the year 2003. After that, the level of stock market concentration was eliminated gradually. Figure 8 in the paper from Schlingemann & Stulz (2021) plots the time series of percentages of Top-1 and Top-3 companies' market capitalization over the market capitalization of all listed firms in US stock market. Except for the scales, our entire trending plotted for the concentration series in Figure 2 is highly consistent with the evidence shown by Schlingemann & Stulz (2021).

Table 1 below also displays the summary statistics of the four control variables. Combined with the specific values, we can obviously notice that the inflation of United States measured by GDP price deflator was at a generally high level at the beginning of the sample period, especially in the late 1970s and early 1980s. Comparatively, the rate has stayed relatively stable, averaging at 0.7703. In terms of GDP growth rate, some years showed negative growths in the US gross domestic product. The growth rate reached the bottom at -3.99 in the second quarter of 2009, and at that moment US and the rest of the world were experiencing the 2008 - 2009 global financial crisis, which is even considered as the deepest recession since World War II. In the 1980s and 1990s, when emergent business activities were expanding, the market's total number of publicly traded companies reached its zenith at 9187 listed firms in the fourth quarter of 1997. In contrast, a large number of corporations delisted from the stock market over the past two decades, reflecting the general downward trend throughout that time period. And the huge variations shown in the number of listed firms capture the one dimension of US stock market's developments, and this can potentially affect the variance components in portfolio diversification potential (Wei & Zhang, 2006).

As a last consideration, the trading volume should also be taken into account based on the previous studies. The proportion of total value of stocks traded on GDP, reported in the last row of Table 1, varies between 321.12 and 7.97 during the 45-year sample period, and there are also quite some variations in this indicator, as can be noticed from the standard deviation, which is around 96 as reported in the third column.

Table 1

Summary Statistics of all the variables

This table records the summary statistics of the main variables under examination for the entire sample period from 1975 to 2019 on a quarterly basis, and both explanatory and control variables are included. Specifically, the last four columns present the information about the average values, standard deviations, the maximums and minimums for each variable contained.

Variables	Mean	Std. Dev.	Maximum	Minimum
NV	0.0561	0.0443	0.3088	0.0045
Div	32.6698	33.9884	223.1463	3.2382
MC_Top5	0.1015	0.0378	0.2032	0.0636
MC_Top10	0.1535	0.0451	0.2749	0.1054
Inflation, GDP price deflator (quarterly)	0.7703	0.55322	2.6330	-0.1747
GDP growth (quarterly %)	2.7524	2.0693	8.5783	-3.9956
Listed companies, total (quarterly)	6817.9060	1112.5290	9187.0000	4778.0000
Stocks traded, total value (quarterly % of GDP)	120.4558	96.0269	321.1200	7.9700

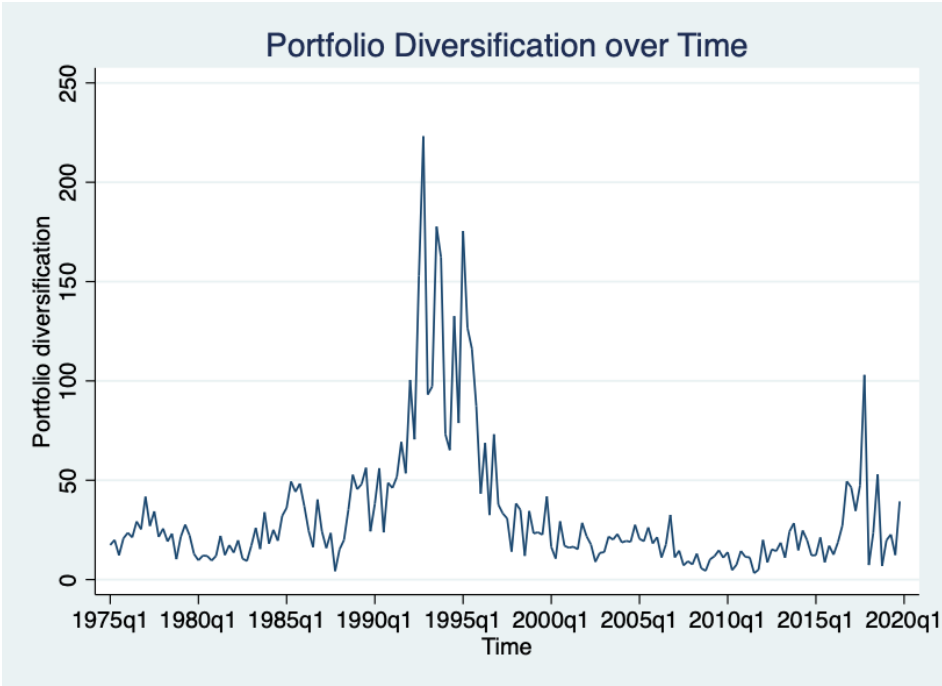

Figure 1. Portfolio Diversification Potential Over Time

This figure plots the changes in portfolio diversification potential series across the sample period. On the y-axis, the degree for the inverse of normalized portfolio variance is presented. The portfolio diversification potential is calculated based on the data of value-weighted market portfolio.

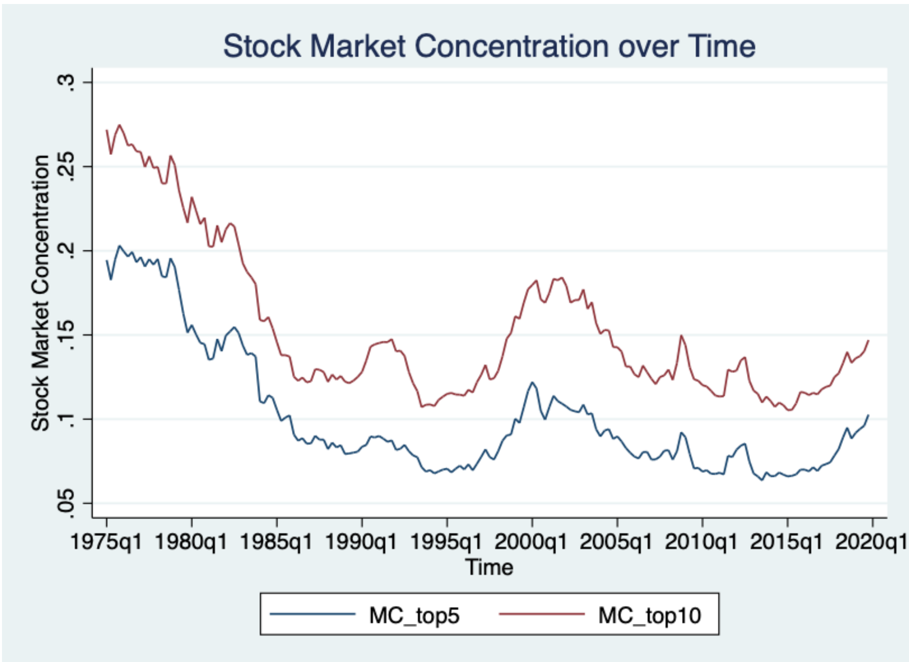

Figure 2. Stock Market Concentration Over Time

This graph illustrates the time series of stock market concentration, measured by the sum of stock market capitalizations of the largest five (ten) companies divided by the total stock market capitalization of US domestic stock exchanges. In the figure, the blue line represents the values computed using the top five firms, and the red line is for the largest ten companies.

3.4.2 US Business Cycle Expansions and Contractions

Data regarding the historical business recessions is collected from the National Bureau of Economic Research (NBER) Database, which provides detailed information about the business cycle reference dates in US history. The specific months and quarters of each peak and trough during our sample period are listed in Table 2 below. Following the standards of business cycle dating in NBER, the recessions are identified as the period between a peak of economic activity and its subsequent trough. During these recessions, a severe decline in economic activity that is widespread across the economy is involved and lasts for more than a few months.

In Table 2 below, the first recession captured is in the 1973 oil crisis and stag inflation recession resulting from stock market crash in 1973-1975 (Fuller, 2006). In the early phase of the 1980s, recessions appeared due to the tight monetary policy in US in order to control inflation. After a lengthy peacetime expansion, the 1990 oil price shock and debt accumulation in the 1980s triggered a brief recession at the beginning of the 1990s (Walsh, 1993). Moreover, the dot-com bubble and the 2008-2009 global financial crisis are both documented in the last two rows in Table 2.

Table 2

Time of US Business Cycle Expansions and Contractions

Peak	Trough
November 1973 (1973Q4)	March 1975 (1975Q1)
January 1980 (1980Q1)	July 1980 (1980Q3)
July 1981 (1981Q3)	November 1982 (1982Q4)
July 1990 (1990Q3)	March 1991 (1991Q1)
March 2001 (2001Q1)	November 2001 (2001Q4)
December 2007 (2007Q4)	June 2009 (2009Q2)

4 Results

In this paper, we examine the potential effects of stock market concentration on portfolio diversification potential. To identify the effects, the ordinary least squares regressions are carried out to test the existence of this unclear relationship, with different macroeconomic factors controlled. In particular, we also analyze this association on different time bases by testing on six separate time lags. Besides that, components of portfolio diversification potential are also tested separately to further explain the results from main analysis. Overall, we begin with the graphical and trend analyses on time-series trends of different variance components, supplementing the study of portfolio diversification potential. The main theme covered in this section is the discussion of results of the analyses above. The first subsection is about the time variation of market volatility based on value-weighted market portfolio. Likewise, the time variation in volatility on the firm-level is also highlighted in the second subsection. Lastly, we discuss the results of the main regressions and component testing and then elaborate the economic meanings behind these outcomes.

4.1 Market-level Volatility

We first plot the time variations of market volatility on a quarterly frequency, which are computed as the variance of value-weighted market portfolio. In our research, the NYSE/AMEX/ Nasdaq composite index is selected as the targeted market portfolio and its corresponding returns are retrieved on a daily frequency. According to the Methodology section, the market portfolio variance is determined as the denominator of portfolio diversification potential.

Figure 3 below plots the changes in the level of market portfolio variance during the sample period, estimated quarterly with daily data. With respect to the specific values, we can see that the market volatility is particularly high in the late 1980s for the value-weighted market portfolio, which is consistent with the evidence shown by Campbell et al. (2001). The stock market crash that occurred in October 1987 led to the enormous spike in market volatility, reaching the second-highest level at 0.0009 in the fourth quarter of that year. Yet, after this cutoff, the market volatility plunge dramatically. From this graph, it can be seen that by far the highest volatility is shown in the last quarter of 2008 at 0.0018, about roughly two times as high as the second highest value. The plot also depicts the NBER-dated recessions shaded in gray. Thus, the global financial crisis in 2008–2009, which is regarded as the most severe recession since the Great Depression, might be held responsible for this peak in market volatility. A casual look at the market volatility and the recession dates can imply that market volatility rises during macroeconomic downturns. During the dot-com bubble around 2000, the associated market volatility also displayed large fluctuations with spikes. More generally, the level of market portfolio variance remains relatively stable over time, as we can observe from the figure. Except for the separate volatility surges during recessions, the figure does not exhibit any visible and huge upward or downward slope. And this also reconciles with evidence from the previous research (Campbell et al., 2001).

To better identify the trends across time, we proceed to analyze the market volatility series in levels by testing for structural breaks and linear trends with specific values. We report the outcomes in detail for these two tests in Table 3. The top panel shows the results of structural breaks for the market-level volatility series. The estimated break date obtained is the third quarter in 2008 and it is also marked as the red line shown in Figure 3. Both supremum Wald statistics and the p-value prove the strong statistical significance of this testing and reject the null hypothesis of no structural breaks. After finding the structural break in the volatility series, we can test the difference in linear trends before and after this sharp turning point. Panel B subsequently presents the trend coefficients from the simple OLS regression of the market volatility on time, based on the equation of trend analysis described in Data & Methodology section. Consider first the period from 1975Q1 to 2008Q2, the trend coefficient shows a small, positive, and statistically significant slope for the market portfolio variance. This finding is also in accord with the positive coefficients found by Campbell et al. (2001) for the linear trend of market volatility over the same period. Contrarily, after the structural break in 2008Q3, the trend in market volatility becomes negative and larger in absolute terms, but statistically insignificant. Although the trend from 2008Q4 to 2019Q4 is slightly larger than the one prior to the break, the magnitudes of both trends remain small and stable across time, confirming the visual evidence from Figure 3. Prior to 2008Q3, the trend is slightly upward, whereas it turns relatively downward after 2008Q3. The slightly distinct slopes shown in both visual evidence and trend regressions prove the presence of structural break in the market volatility series at the third quarter of 2008. Although the level of market portfolio variance is declining modestly after the structural break, the general trend has remained steady in the recent twenty years.

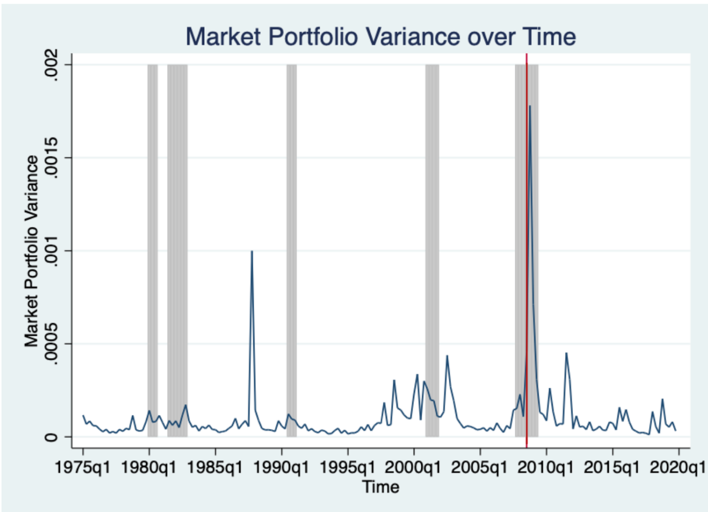

Figure 3. Market Volatility Over Time

The figure shows the time variations in variance of the daily returns for market index, calculated as the denominator of the portfolio diversification potential formula. The time period runs from 1975 to 2019 on a quarterly basis. The line captures the trend of variance of value-weighted market portfolio. NBER-dated recessions from Table 2 are shaded in gray to indicate the cyclical movements in volatility. The red line marks the date of identified structural break.

Table 3

Testing for Structural Breaks and Linear Trends of Market Portfolio Variance (Market-level Volatility)

This table reports the regression estimates of structural break testing and linear trend testing. Panel A presents the estimated break date as well as relevant statistics. Panel B reports the linear trend testing outputs on the volatility series of market portfolio variance (market-level volatility). The sample lasts for a 45-year period from 1975 to 2019. All the variables are on the quarterly frequency. The t-statistics in parentheses are based on robust standard errors.

Panel A: Structural Break Testing - Market Portfolio Variance (Market-level Volatility)

Estimated Break Date	2008Q3
Supremum Wald Statistics	33.2781
P-value	0.0000
Observations	180

Panel B: Linear Trend Testing - Market Portfolio Variance (Market-level Volatility)

	Structural break **2008Q3**	
	1975Q1 – 2008Q2	2008Q4 – 2019Q4
t (*10^4)	0.0037	-0.0545
	(2.35)	(-1.35)
ave_var_{t-1}	0.2566	0.3797
	(2.21)	(3.63)
Constant	0.0000	0.0013
	(0.84)	(1.41)
Observations	134	45
R-squared	0.0956	0.3006

4.2 Firm-level Volatility

A fast-growing literature emphasizes the role of idiosyncratic risk of stock returns. In line with Campbell et al. (2001), the firm-level volatility is calculated as the average variance of individual stocks contained in the sample, and Figure 4 clearly plots its evolution across the sample period. The first striking feature in the graph is that the variance here is, on average, higher than the market volatility depicted in the previous subsection. This is a manifestation that the idiosyncratic risk accounts for a huge component in the total volatility of an average firm. In detail, the upward trend reached the second highest at 0.0054 in the fourth quarter of 1998. However, following 1998, the firm-specific volatility begins a slow but steady decline, only to start climbing again and reach its highest point at 0.0078 during the 2008-2009 global financial crisis. Similar to the tests on market volatility series, we conduct two research testing for the structural breaks and linear trends in the firm-level volatility to better capture the time variations. Table 4 contains two panels: the first reports the estimated structural break date and corresponding statistics showing the level of significance; Panel B lists the regression estimates for the linear trends in average variance of individual stocks. As marked as the red line in Figure 4, the second quarter of 2003 is identified as the structural break in the firm-level volatility series with strong statistical significance. Besides that, it can be distinctly witnessed that the volatility in Figure 4 trends upward throughout the first half of the sample period, although Figure 3 of market volatility does not demonstrate such an apparent trend. And this can also be confirmed with the trend coefficient in Panel B of Table 4. For the period prior to 2003Q2 (the series on the left side of the red line), the trend coefficient for firm-level volatility is 0.1230 with statistical significance, which is approximately four times higher than the trend coefficient for market volatility series before the structural break. This gives the implication that the volatility on a firm level is increasing much more rapidly than that on a market level in the 20th century, which is consistent with the results from Campbell et al. (2001) under the same examining period. Regarding the series after the break date, the level of average variance of individual stocks shows a small, downward but insignificant trend. Hence, during the recent twenty years, both market volatility and firm-level volatility are facing the declining trends, but both tendencies are small and statistically insignificant.

When factoring in the areas that are shaded, the firm-level volatility is higher during these recessions that have been dated by the NBER according to Table 2, which indicates the significant effects of economic downturns. The overlapping sample period of this study and the research of Campbell et al. (2001) ranges from 1975 to 1997. All features captured during this period in our paper are consistent with the findings of Campbell et al. (2001). In general, the US stocks have become less volatile on the firm-level basis averagely in the recent twenty years, compared with the late 20th century.

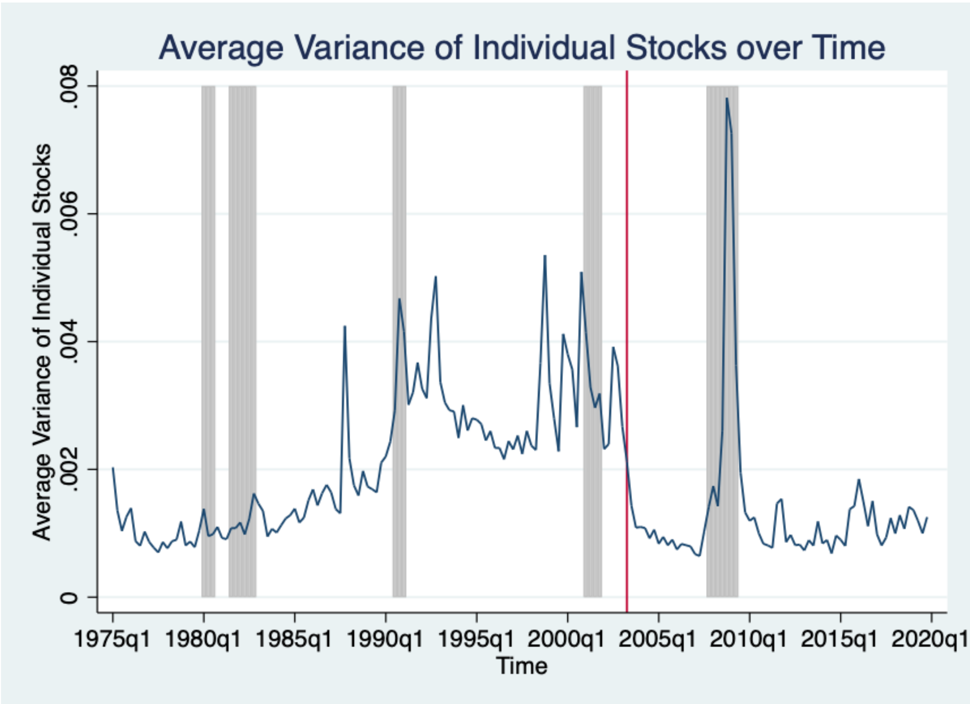

Figure 4. Firm-level Volatility Over Time

The figure shows the time variations in average variance of individual stocks contained in stock market, calculated as the numerator of the portfolio diversification potential formula. The time period runs from 1975 to 2019 on a quarterly basis. NBER-dated recessions from Table 2 are shaded in gray to indicate the cyclical movements in volatility. The red line marks the date of identified structural break.

Table 4

Testing for Structural Breaks and Linear Trends of Average Variance of Individual Stocks (Firm-level Volatility)

This table reports the regression estimates of structural break testing and linear trend testing. Panel A presents the estimated break date as well as relevant statistics. Panel B reports the linear trend testing outputs on the volatility series of average variance of individual stocks (firm-level volatility). The sample lasts for a 45-year period from 1975 to 2019. All the variables are on the quarterly frequency. The t-statistics in parentheses are based on robust standard errors.

Panel A: Structural Break Testing - Average Variance of Individual Stocks (Firm-level Volatility)

Estimated Break Date	2003Q2
Supremum Wald Statistics	130.3747
P-value	0.0000
Observations	180

Panel B: Linear Trend Testing - Average Variance of Individual Stocks (Firm-level Volatility)

	Structural break	
	2003Q2	
	1975Q1 – 2003Q1	2003Q3 – 2019Q4
$t*10^4$	0.1230	-0.0069
	(3.70)	(-0.24)
ave_var_{t-1}	0.5389	0.7312
	(4.21)	(4.21)
Constant	0.0005	0.0005
	(0.80)	(0.80)
Observations	113	66
R-squared	0.7140	0.5398

4.3 Regression Results

4.3.1 Regressions of Portfolio Diversification Potential on Stock Market Concentration

The empirical question of this analysis is to answer how the stock market concentration affects portfolio diversification potential. The findings of regression analyses with daily US stock data from 1975 to 2019 are shown in Tables 5. Table 6 parallels Table 5 but uses different time lags. Table 5 specifically exhibits the outcomes utilizing lagged market concentration by 5, 4 and 3 years. Likewise, regression results based on 2- and 1-year-lagged plus contemporary market concentration are displayed in Table 6. The level of portfolio diversification, which is the inverse of the normalized variance of value-weighted market portfolio, is investigated as the dependent variable in each table. In terms of the independent variables, the lagged stock market concentrations measured with both the largest five and the largest ten corporations are tested, respectively. The two tables subsequently list the regression outcomes for the various control variables.

Using the full sample, we find out a huge and negative relation between the two variables of interest in Table 5, especially shown in the 5-year and 4-year lags. According to the coefficient of $Mkt\_Con\_Top5_{t-5}$, we notice that with one standard deviation increase (0.0378) in the level of stock market concentration by the top five companies, the portfolio diversification potential of value-weighted market portfolio five years later will decrease by around 18.1981 percentage points (0.0378×-481.4316). And this relation is highly statistically significant at 1% level. A similar scale and statistical significance also appear under the 4-year-lagged stock market concentration. With 3-year lags, documented in the last two columns, the relation revealed remains negative but is relatively less significant. Using the stock market concentration measured with top 10 firms, the coefficients are approximately half of those with top 5 firms but still display the same signs. Based on the measurement of portfolio diversification potential, we can also indicate that the negative signs shown in coefficients also reflects the increase in normalized portfolio variance, meaning a more concentrated portfolio. Therefore, our results in Table 5 have important the economic implication that the stock market concentration is strongly negatively associated with the portfolio diversification in the future three, four and five years. In other words, the more concentrated the stock market is, the less diversification potential a portfolio will have. And this finding is consistent in the long run with our hypothesis.

Moving to Table 6, comparable outcomes are revealed in a shorter time frame. The coefficients of market concentration are all large and negative. And this relation is mostly statistically significant at 1% level as reported from column (3) to (6). Taking $Mkt\_Con\_Top10_{q,t-1}$ as the example, the coefficient shown in column (4) implies that an increase of one standard deviation (0.0451) in the level of stock market concentration by the top ten firms is associated with a decline in the portfolio diversification potential in one year by around 8.1545 percentage points (0.0451 × -180.8085) on the 1% significance level. Hence, based on the outputs in Table 6, we uncover that over the short haul, the increased stock market concentration will also cause the corresponding decline in portfolio diversification potential. In the short term, the conjecture is again clearly confirmed, economically showing that the more concentrated the stock market is, the less diversification potential will be for the portfolio in the short run. And strong statistical significance is also observed in this evidence.

When examining the temporal effects, it is noticeable that the scale of the effects in absolute terms we capture is consistently diminishing as the concurrent year approaches. For the five-year lag, the scale of coefficient on the level of stock market concentration by top five firms is 481.4316 in absolute terms as reported in the column (1) of Table 5. And it gradually drops to 311.6857 in absolute terms for the 2-year-lagged values in column (1) of Table 6. Although there is a jump under the 1-year lag, the increase is rather small, and the scale continues to decrease when approaching the contemporary year t. We also replace Mkt_Con_Top5 with Mkt_Con_Top10 and find a relatively smaller but similar trend for the time variations. More importantly, the statistical significances also show some slight variations across different time lags. The negative relation is highly statistically significant at 1% level for 5- and 4-year lags. However, it shows weaker significance for portfolio diversification potential in two and three years later at 5% and 10% levels, as presented by column (5) and (6) in Table 5 plus column (1) and (2) in Table 6. When it is closer to the contemporary year, the relation appears to be highly significant at 1% level again under the 1-year lag and concurrent year. Taken together, what emerges from the results reported here is that the increased stock market concentration will decrease the portfolio diversification potential both in the long run and short run. Specifically, the magnitude of this negative effect is stronger in the long term. And portfolio diversification potential exhibits relatively smaller short-run and immediate reactions towards the increased concentration in stock market. From the statistical perspective, although modest variations exist, the significance of revealed relation between stock market concentration and portfolio diversification potential is strong in general across time.

In addition to the explanatory variables, the two tables also provide the regression statistics regarding different macroeconomic control variables. Interestingly, most of the inflation rates reflect the negative but small associations with the diversification potential of value-weighted market portfolio. And all the results lose statistical significance in both long and short term. This is in line with the findings of Schwert (1989) that inflation is linked with the volatility of short- and long-run asset returns but the relation is not very strong and close. Regarding the variable capturing the cyclical behavior of volatility, GDP growth is positively correlated with the portfolio diversification for the long haul. While in Table 6, more negative correlations are displayed in the short run. Same with the variable of inflation, no statistical significance is observed across years. The correlation reported for the short haul is consistent with the existing finding that the firm-level volatility embodies a negative correlation with GDP growth (Campbell et al., 2001). In addition, looking at the total number of listed firms, the relation is rather small and positive, meaning that more listed firms included in the stock market is associated with higher diversification potential of portfolios. The coefficients are statistically significant when approaching the contemporary year. In general, the outcome is consistent with that of Wei & Zhang (2006), who found out that the increase in number of listed firms contributes to the upward trend in firm-level volatility. Based on our outputs, the last control variable $Stocks\_Traded_t$ shows a highly strong and negative relation with portfolio diversification, reconfirming the significant relation between trading volume and stock market volatility found by (Schwert, 1989).

4.3.2 Regressions for Component Testing

According to the outcomes of the main regressions shown in the last subsection, we can generate the finding that the stock market concentration is significantly negatively correlated with the diversification potential of portfolios. In this subsection, we further examine the relation between the components of diversification potential and the stock market concentration. Specifically, we test whether a more concentrated stock market affects the market portfolio variance and the average variance of individual stocks, which are the denominator and numerator of the dependent variables in main regressions. This experiment is an essential step in exploring why there is such a significant relation existing between stock market concentration and portfolio diversification potential.

Table 5

Regressions of portfolio diversification potential on stock market concentration at t-5, t-4 and t-3

This table reports the relation between portfolio diversification potential and stock market concentration by regressing them on the macroeconomic control variables. The sample lasts for a 45-year period from 1975 to 2019 on a quarterly basis. The dependent variables are the inverse of normalized portfolio variance of value-weighted market portfolio. All the variables are on the quarterly frequency. The t-statistics in parentheses are based on robust standard errors. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
$Mkt\_Con\_Top5_{q,t-5}$	-481.4316***					
	(-4.10)					
$Mkt\_Con\_Top10_{q,t-5}$		-333.7163***				
		(-3.51)				
$Mkt\_Con\_Top5_{q,t-4}$			-469.6965***			
			(-3.53)			
$Mkt\_Con\_Top10_{q,t-4}$				-270.8326*		
				(-2.46)		
$Mkt\_Con\_Top5_{q,t-3}$					-417.6879**	
					(-2.93)	
$Mkt\_Con\_Top10_{q,t-3}$						-204.8910
						(-1.90)
$Inflation_{q,t}$	1.0798	0.0186	-0.1998	-3.5932	-1.9065	-6.5611
	(0.32)	(0.01)	(-0.06)	(-1.09)	(-0.53)	(-1.86)
$GDP\_Growth_{q,t}$	0.4075	0.5670	0.3015	0.4579	0.6536	0.3464
	(0.63)	(0.78)	(0.44)	(0.54)	(0.71)	(0.34)
$Listed\_Firms_{q,t}$	0.0029	0.0036	0.0024	0.0045	0.0028	0.0059
	(0.84)	(0.99)	(0.60)	(0.96)	(0.61)	(1.15)
$Stocks\_Traded_{q,t}$	-0.2668***	-0.2377***	-0.2608***	-0.2310***	-0.2475***	-0.2249***
	(-6.73)	(-6.79)	(-6.68)	(-6.70)	(-6.66)	(-6.73)
Constant	97.5829**	91.2619*	98.8447*	76.3103	88.2544	57.1184
	(2.68)	(2.32)	(2.35)	(1.58)	(1.94)	(1.13)
Observations	180	180	180	180	180	180
R-squared	0.3832	0.3693	0.3705	0.3434	0.3514	0.3265

T-statistics in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 6

Regressions of portfolio diversification potential on stock market concentration at t-2, t-1 and t

This table reports the relation between portfolio diversification potential and stock market concentration by regressing them on the macroeconomic control variables. The sample lasts for a 45-year period from 1975 to 2019 on a quarterly basis. The dependent variables are the inverse of normalized portfolio variance of value-weighted market portfolio. All the variables are on the quarterly frequency. The t-statistics in parentheses are based on robust standard errors. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
$Mkt\_Con\_Top5_{q,t-2}$	-311.6857**					
	(-2.99)					
$Mkt\_Con\_Top10_{q,t-2}$		-149.8541*				
		(-2.36)				
$Mkt\_Con\_Top5_{q,t-1}$			-327.2638***			
			(-3.63)			
$Mkt\_Con\_Top10_{q,t-1}$				-180.8085***		
				(-3.36)		
$Mkt\_Con\_Top5_{q,t}$					-318.6630***	
					(-3.79)	
$Mkt\_Con\_Top10_{q,t}$						-239.5535***
						(-3.82)
$Inflation_{q,t}$	-2.0997	-6.2707	-2.5859	-5.0159	-2.5564	-1.7298
	(-0.53)	(-1.52)	(-0.62)	(-1.13)	(-0.60)	(-0.39)
$GDP\_Growth_{q,t}$	0.2152	-0.1476	0.2829	-0.2265	0.0369	-0.4045
	(0.27)	(-0.19)	(0.37)	(-0.32)	(0.05)	(-0.59)
$Listed\_Firms_{q,t}$	0.0061	0.0083*	0.0063*	0.0084**	0.0078**	0.0090**
	(1.64)	(2.32)	(2.02)	(2.91)	(2.71)	(3.30)
$Stocks\_Traded_{q,t}$	-0.2305***	-0.2175***	-0.2281***	-0.2167***	-0.2239***	-0.2149***
	(-6.83)	(-6.91)	(-6.88)	(-6.90)	(-6.93)	(-6.97)
Constant	53.2989	31.2591	52.0392	34.1388	40.8085	36.5112
	(1.55)	(0.98)	(1.79)	(0.173)	(1.58)	(1.50)
Observations	180	180	180	180	180	180
R-squared	0.3320	0.3180	0.3410	0.3254	0.3469	0.3445

T-statistics in parentheses

*** p<0.01, ** p<0.05, * p<0.1

To identify the potential relations, we break down the measurement of diversification potential and replace the dependent variables of the main regressions with the corresponding components. Table 7 and 8 in Appendix report the outputs of regressing average variance of individual stocks on stock market concentration while controlling for the macroeconomic variables. The difference between these two tables is the different time lags applied. Table 9 and 10 in the Appendix mirror the Table 7 and 8 but use the market portfolio variance as the dependent variable instead. All the independent variables and control variables are identical to the main regressions.

It can be obviously seen that in Table 7, no statistical significance is displayed for the coefficients obtained and all the scales in absolute terms are relatively small and close to zero. In contrast, the relation exhibits strong significance in Table 8. In particular, the statistical significance becomes stronger when approaching the contemporary year and reaches 1% level under 1-year lags and concurrent year. Besides, the magnitudes of these coefficients are relatively greater than those in Table 7 and also offer economic implications. Taking column (3) of Table 8 as example, a one standard deviation increase (0.0378) in the level of stock market concentration by the top five corporates predicts a drop of approximately 0.0005 percentage points (0.0378×-0.0141) in the average variance of individual stocks in one year. Although the coefficients' signs change across time, the number and statistical significance of negative correlations are generally both higher than those of positive relations. Turning to the effects on market portfolio variance, small and positive correlations are captured in Table 9 and 10 for all time lags. This relation is mostly significant at 10% level and becomes strongest at 5% level for both Top5 and Top 10 in the contemporary year shown in Table 10. The magnitudes of the effects in absolute terms also provide economic implications. Looking at the column (6) in Table 10 for instance, an increase of one standard deviation (0.0451) in the degree of stock market concentration by the top ten firms leads to an increase in the market portfolio variance by around 0.00005 percentage points

(0.0451×0.0012). Compared with the scales of effects on the average variance of individual stocks, the effects on the market portfolio variance have smaller magnitudes.

Overall, stock market concentration particularly has strong effects on the average variance of individual stocks which is the numerator of the portfolio diversification potential. And the effects are most statistically significant at 1% level in the short term. Additionally, we also find out that stock market has potential influence on the market portfolio variance, which is the denominator of portfolio diversification potential. Unlike the firm-level volatility, the effects on market portfolio variance are weaker but remain at 5% and 10% levels in both long and short runs. Therefore, the strong relations between stock market concentration and the components of diversification can, to some extent, confirm and explain why stock market significantly affects diversification potential. With respect to the specific effects, stock market concentration generally has significantly negative impacts on average variance of individual stocks and significantly positive impacts on the market portfolio variance. Besides that, the magnitudes of effects on average variance of individual stocks are relatively greater than those on market portfolio variance. Combined with the measurement of portfolio diversification potential, the negative impact on the numerator is larger than the positive impacts on the denominator. Hence, the negative correlation between stock market concentration and portfolio diversification potential can be explained from the perspectives of both the sign and magnitude of the effects on portfolio diversification's components.

5 Robustness Check

In this section, we display the two additional tests to further substantiate the link between stock market concentration and portfolio diversification potential. Since there is a growing body of literature on how to measure the level of market concentration, we wish to first identify whether the measurement of stock market concentration affects the findings. Therefore, an alternative measurement of stock market concentration is applied and tested in the regression models. Next, instead of using the value-weighted market portfolio in the main study, we compute the market portfolio variance using the equal-weighted market portfolio. To further evaluate the robustness of our results, we conduct identical regressions to the main regressions but use the diversification potential of an equal-weighted market portfolio as the dependent variable.

6.1 Measuring Stock Market Concentration with Herfindahl-Hirschman Index

Following the methodology from Grullon et al. (2019), an alternative measurement of market concentration, Herfindahl-Hirschman Index (HHI), is adopted to conduct this robustness check. This is calculated by summing up the market shares of all firms included in the stock market:

$$HHI_{q,t} = s_{1,q,t}^2 + s_{2,q,t}^2 + s_{3,q,t}^2 + \cdots + s_{n,q,t}^2$$

where $s_{i,q,t}^2$ delivers the market share of firm i at the quarter q in year t, and it is considered as the proportion of firm i's market capitalization on the total market capitalization of the stock market. Next, the independent variables are replaced with the lagged Herfindahl-Hirschman Index and similar regressions as presented in the methodology section are tested:

$$\begin{aligned} Div_{q,t} = \beta_0 + \beta_1 HHI_{q,t-j} + \gamma_1 Inflation_{q,t} + \gamma_2 GDP\_Growth_{q,t} \\ + \gamma_3 Listed\_Firms_{q,t} + \gamma_4 Stocks\_Traded_{q,t} + \epsilon \end{aligned}$$

To address the concern, Table 11 below depicts the regression estimates similar to Table 5 but regressed by the Herfindahl-Hirschman Index lagged by five, four and three years, respectively. Table 12 mirrors the regressions in Table 11 but uses the 2- and 1-year lags plus the value on the concurrent year. As a result of the different measuring procedure, the coefficients in these two tables have a much greater scale than those in the main results. Nonetheless, the time variations of the coefficients and statistical significance are comparable to the earlier findings. Similar to the results in Table 5, the coefficient shown in the first column of Table 11 is also significantly negative at 1% level. Both the magnitudes in absolute terms and statistical significances of the coefficients decline over time in Table 11. Eventually, in Table 12, the magnitudes reach the minimum under the 2-year lagged HHI. Although there is a small increase under the 1-year lag, the magnitude keeps decreasing until arriving the contemporary year. Same with our main results, the statistical significance also returns to a highly strong degree in the contemporary year at 1% level. The coefficients in 4-year, 3-year and 2-year lags show slightly weaker significance but remain statistically significant at 5% and 10% levels.

In general, all these regression estimates give the economic implications that the effect of increased stock market concentration is significantly negative on portfolio diversification potential for both long and short hauls. Namely, the level of diversification potential of portfolios will decline when the stock market turns more concentrated. Overall, we find consistent evidence that supports the stock market concentration is negatively correlated with portfolio diversification, using a different measurement of stock market concentration.

6.2 Computing Portfolio Diversification Potential with Equal-weighted Market Portfolio

As we mentioned in the Data & Methodology section, returns of the value-weighted market composite index are applied in our main analysis to calculate the market portfolio variance and further the portfolio diversification potential.
However, the returns of market portfolio can also be formed by giving equal importance to the stock of each firm that makes up the market index. Therefore, we base our calculation of portfolio diversification potential on the data from equally weighted market portfolio and investigate whether our findings from main results remain valid under the equal-weighted market portfolio. To be specific, we first calculate the market portfolio variance with equal-weighted returns of the market portfolio. Then the variable portfolio diversification can also be generated by replacing the previous denominator with this new market portfolio variance. Lastly, regressions will be tested on it and lagged stock market concentration while controlling for the macroeconomic variables.

Table 13 and 14 report the regression estimates with different time lags. It is evident that the testing results remain unchanged: all the coefficients are negative and statistically significant in both long and short runs. The magnitudes in absolute terms of the coefficients trend downward in the long term and reach the minimum level at 2-year lags. Despite the small increase, the magnitudes continue to drop in the contemporary year. Interestingly, the scales of the negative effects shown are relatively larger than those in the main regression. Ultimately, our hypothesis and the main study's findings are robustly supported by the significantly negative but diminishing effects of stock market concentration on portfolio diversification potential shown in this testing.

6 Conclusions & Limitations

As market competition is getting fierce, corporates which are more productive and innovative gradually stand out in this competition. The influence of these leading companies even touches upon the real economy. The financial system, a key player in the economy activity, has the fundamental duty is to promote the effective deployment of capital and economic resources (Bodie & Merton, 1998). Therefore, with more successful firms arising, more capital and resources are distributed to these leading firms (Bae et al., 2020). In this paper, we embark on a different aspect of financial market, the stock market concentration. To dive in, we link it to the portfolio diversification potential which is inspired by the increasing volatility recorded in both aggregate market and firm-specific levels. In short, this study set out to assess the impacts of stock market concentration on the portfolio diversification potential. With daily stock data from United States, this paper has identified the evidence that the stock market concentration is significantly negatively associated with the portfolio diversification both in the long and short runs, which is in line with our proposed hypothesis. Meanwhile, the magnitudes of these time effects are relatively large and persistently become smaller as the time approaches the concurrent year. Economically speaking, the negative effects of stock market concentration will amplify on the portfolio diversification potential in the longer term. Moreover, the results regarding control variables are mostly consistent with the prior literature. Component testing is also performed, and the outcomes obtained prove the significant relations between stock market concentration and components of diversification potential, explaining underlyingly why the significant correlation exists between the two main variables. For the robustness of our results, both robustness checks reconfirm our findings and hypothesis. In addition, this study also provides the updated time variations in both stock market volatility and firm-level volatility. During the recent twenty years, small, negative but statistically insignificant trends have been found for both market-level and firm-level volatility.

In general, deeper insights are provided by this paper into the rapidly expanding field of market concentration. These findings contribute to the knowledge gap on how stock market concentration correlates with portfolio diversification potential. Besides, the research also offers some new implications towards individual investors and portfolio managers when investing in a relatively concentrated stock market. Lastly, we also update the time variations of market and firm-level volatility captured from Campbell et al. (2001) by extending the sample period by almost thirty years. However, our results are still restricted by the sample size. Also, only stock market in United States is under investigation in this paper. Due to the diversity among different national stock markets and real economies, what is needed for the future studies is to investigate this topic in a more international setting, capturing overall developments of different stock markets and the corresponding influence plus implications on international portfolio diversification. Last but not least, the results of our research point to the necessity for analysts and private investors to incorporate various forms of marginal analysis into their portfolio selection models.

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8 Appendix

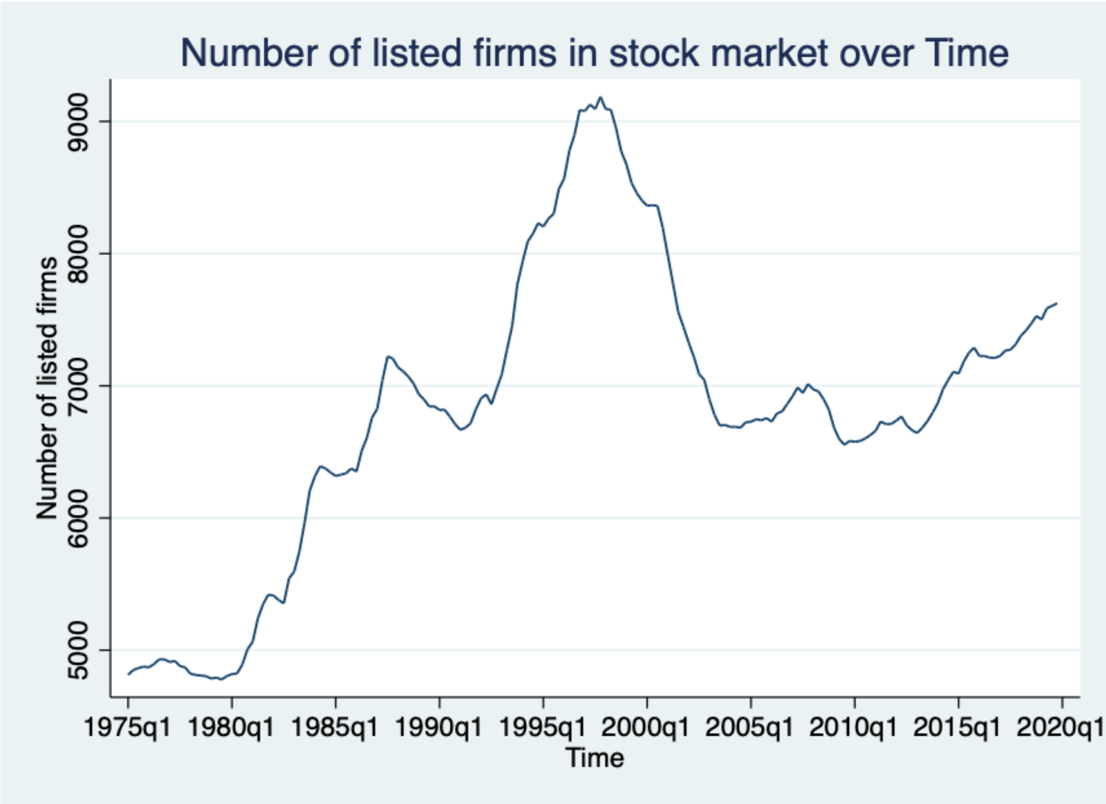

Figure 5 Number of Listed Firms in US Stock Market over Time

The figure shows the time variations in the number of listed firms in US stock market throughout the sample period. The time period runs from 1975 to 2019 on a quarterly basis. The y-axis denotes the number of listed firms and quarterly time series are listed on the x-axis.

Table 7

Regressions of average variance of individual stocks (firm-level volatility) on stock market concentration at t-5, t-4 and t-3

This table reports the relation between average variance of individual stocks (firm-level volatility) and stock market concentration by regressing them on the macroeconomic control variables. The sample lasts for a 45-year period from 1975 to 2019 on a quarterly basis. The dependent variables are the average variance of individual stocks included in the stock market. All the variables are on the quarterly frequency. The t-statistics in parentheses are based on robust standard errors. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
$Mkt\_Con\_Top5_{q,t-5}$	-0.0020					
	(-0.45)					
$Mkt\_Con\_Top10_{q,t-5}$		-0.0033				
		(-0.90)				
$Mkt\_Con\_Top5_{q,t-4}$			-0.0004			
			(-0.09)			
$Mkt\_Con\_Top10_{q,t-4}$				-0.0011		
				(-0.31)		
$Mkt\_Con\_Top5_{q,t-3}$					0.0057	
					(1.16)	
$Mkt\_Con\_Top10_{q,t-3}$						0.0034
						(1.01)
$Inflation_{q,t}$	-0.0001	-0.0000	-2.5859	-5.0159	-2.5564	-1.7298
	(-0.39)	(-0.17)	(-0.57)	(-0.49)	(-1.15)	(-0.96)
$GDP\_Growth_{q,t}$	-0.0002**	-0.0002**	-0.0002**	-0.0002**	-0.0002**	-0.0002**
	(-3.13)	(-2.84)	(-3.25)	(-3.00)	(-3.45)	(-3.42)
$Listed\_Firms_{q,t}$	0.0000***	0.0000***	0.0000***	0.0000***	0.0000***	0.0000***
	(4.79)	(4.65)	(4.47)	(4.43)	(4.92)	(4.95)
$Stocks\_Traded_{q,t}$	-0.0000**	-0.0000**	-0.0000**	-0.0000**	-0.0000**	-0.0000**
	(-2.79)	(-3.20)	(-2.64)	(-3.02)	(-2.28)	(-2.72)
Constant	-0.0008	-0.0002	-0.0012	-0.0009	-0.0027	-0.0025
	(-0.61)	(-0.18)	(-0.80)	(-0.63)	(-1.68)	(-1.56)
Observations	180	180	180	180	180	180
R-squared	0.3227	0.3265	0.3217	0.3221	0.3284	0.3260

T-statistics in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 8

Regressions of average variance of individual stocks (firm-level volatility) on stock market concentration at t-2, t-1 and t

This table reports the relation between average variance of individual stocks (firm-level volatility) and stock market concentration by regressing them on the macroeconomic control variables. The sample lasts for a 45-year period from 1975 to 2019 on a quarterly basis. The dependent variables are the average variance of individual stocks included in the stock market. All the variables are on the quarterly frequency. The t-statistics in parentheses are based on robust standard errors. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
$Mkt\_Con\_Top5_{q,t-2}$	0.0112*					
	(2.51)					
$Mkt\_Con\_Top10_{q,t-2}$		-0.0075**				
		(2.68)				
$Mkt\_Con\_Top5_{q,t-1}$			-0.0141***			
			(3.43)			
$Mkt\_Con\_Top10_{q,t-1}$				-0.0106***		
				(3.96)		
$Mkt\_Con\_Top5_{q,t}$					-0.0135***	
					(3.77)	
$Mkt\_Con\_Top10_{q,t}$						0.0125***
						(4.89)
$Inflation_{q,t}$	-0.0005	-0.0004	-0.0005*	-0.0005*	-0.0005*	-0.0006*
	(-1.89)	(-1.66)	(-2.21)	(-2.25)	(-2.20)	(-2.75)
$GDP\_Growth_{q,t}$	-0.0002***	-0.0002***	-0.0002***	-0.0002***	-0.0002***	-0.0002***
	(-3.81)	(-3.82)	(-4.06)	(-3.98)	(-3.97)	(-3.92)
$Listed\_Firms_{q,t}$	0.0000***	0.0000***	0.0000***	0.0000***	0.0000***	0.0000***
	(6.39)	(6.49)	(7.27)	(7.44)	(7.91)	(8.51)
$Stocks\_Traded_{q,t}$	-0.0000*	-0.0000*	-0.0000*	-0.0000*	-0.0000*	-0.0000*
	(-2.31)	(-2.79)	(-2.47)	(-3.01)	(-2.78)	(-3.32)
Constant	-0.0038**	-0.0037**	-0.0042***	-0.0043***	-0.0037***	-0.0040***
	(-2.73)	(-2.71)	(-3.38)	(-3.48)	(-3.55)	(-4.14)
Observations	180	180	180	180	180	180
R-squared	0.3482	0.3457	0.3730	0.3747	0.3800	0.4044

T-statistics in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 9

Regressions of market portfolio variance (market volatility) on stock market concentration at t-5, t-4 and t-3

This table reports the relation between market portfolio variance (market volatility) and stock market concentration by regressing them on the macroeconomic control variables. The sample lasts for a 45-year period from 1975 to 2019 on a quarterly basis. The dependent variables are the variance of value-weighted market portfolio throughout the sample period. All the variables are on the quarterly frequency. The t-statistics in parentheses are based on robust standard errors. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
$Mkt\_Con\_Top5_{q,t-5}$	0.0018*					
	(2.18)					
$Mkt\_Con\_Top10_{q,t-5}$		0.0012				
		(1.91)				
$Mkt\_Con\_Top5_{q,t-4}$			0.0018*			
			(2.41)			
$Mkt\_Con\_Top10_{q,t-4}$				0.0010*		
				(2.08)		
$Mkt\_Con\_Top5_{q,t-3}$					0.0018**	
					(2.98)	
$Mkt\_Con\_Top10_{q,t-3}$						0.0010*
						(2.35)
$Inflation_{q,t}$	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000
	(-1.32)	(-1.14)	(-1.30)	(-0.96)	(-1.35)	(-0.80)
$GDP\_Growth_{q,t}$	-0.0000*	-0.0000*	-0.0000*	-0.0000*	-0.0000*	-0.0000*
	(-2.24)	(-2.17)	(-2.24)	(-2.16)	(-2.28)	(-2.18)
$Listed\_Firms_{q,t}$	0.0000*	0.0000*	0.0000*	0.0000*	0.0000*	0.0000*
	(2.19)	(2.04)	(2.39)	(2.20)	(3.24)	(2.58)
$Stocks\_Traded_{q,t}$	0.0000**	0.0000**	0.0000**	0.0000**	0.0000**	0.0000**
	(2.78)	(2.62)	(2.85)	(2.61)	(2.87)	(2.64)
Constant	-0.0003	-0.0003	-0.0003*	-0.0003	-0.0004**	-0.0002
	(-1.86)	(-1.66)	(-2.06)	(-1.77)	(-2.79)	(-1.97)
Observations	180	180	180	180	180	180
R-squared	0.1937	0.1832	0.1872	0.1720	0.1836	0.1678

T-statistics in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 10

Regressions of market portfolio variance (market volatility) on stock market concentration at t-2, t-1 and t

This table reports the relation between market portfolio variance (market volatility) and stock market concentration by regressing them on the macroeconomic control variables. The sample lasts for a 45-year period from 1975 to 2019 on a quarterly basis. The dependent variables are the variance of value-weighted market portfolio throughout the sample period. All the variables are on the quarterly frequency. The t-statistics in parentheses are based on robust standard errors. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
$Mkt\_Con\_Top5_{q,t-2}$	0.0015*					
	(2.35)					
$Mkt\_Con\_Top10_{q,t-2}$		0.0007				
		(1.87)				
$Mkt\_Con\_Top5_{q,t-1}$			0.0015*			
			(2.48)			
$Mkt\_Con\_Top10_{q,t-1}$				0.0008		
				(1.97)		
$Mkt\_Con\_Top5_{q,t}$					0.0016**	
					(2.77)	
$Mkt\_Con\_Top10_{q,t}$						0.0012**
						(2.88)
$Inflation_{q,t}$	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000
	(-1.30)	(-0.82)	(-1.24)	(-0.92)	(-1.37)	(-1.49)
$GDP\_Growth_{q,t}$	-0.0000*	-0.0000*	-0.0000*	-0.0000*	-0.0000*	-0.0000*
	(-2.18)	(-2.12)	(-2.22)	(-2.16)	(-2.22)	(-2.15)
$Listed\_Firms_{q,t}$	0.0000**	0.0000	0.0000**	0.0000	0.0000**	0.0000*
	(2.67)	(1.81)	(2.79)	(1.55)	(2.73)	(2.07)
$Stocks\_Traded_{q,t}$	0.0000**	0.0000**	0.0000**	0.0000**	0.0000**	0.0000**
	(2.73)	(2.62)	(2.75)	(2.67)	(2.77)	(2.71)
Constant	-0.0002*	-0.0001	-0.0002*	-0.0001	-0.0002*	-0.0002
	(-2.05)	(-1.09)	(-1.99)	(-0.91)	(-2.16)	(-1.92)
Observations	180	180	180	180	180	180
R-squared	0.1747	0.1609	0.1783	0.1632	0.1898	0.1875

T-statistics in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 11

Regressions of portfolio diversification potential on stock market concentration (HHI) at t-5, t-4 and t-3

This table reports the relation between portfolio diversification potential and stock market concentration by regressing them on the macroeconomic control variables. The sample lasts for a 45-year period from 1975 to 2019 on a quarterly basis. The dependent variables are the inverse of normalized portfolio variance of value-weighted market portfolio. Independent variables in this figure are lagged Herfindahl-Hirschman Index. All the variables are on the quarterly frequency. The t-statistics in parentheses are based on robust standard errors. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)
$HHI_{q,t-5}$	-5664.3330***		
	(-4.05)		
$HHI_{q,t-4}$		-4837.1840**	
		(-3.07)	
$HHI_{q,t-3}$			-3973.1600*
			(-2.45)
$Inflation_{q,t}$	1.4943	-2.1430	-4.7375
	(0.42)	(-0.64)	(-1.31)
$GDP\_Growth_{q,t}$	0.5918	0.3253	0.5072
	(0.85)	(0.43)	(0.53)
$Listed\_Firms_{q,t}$	0.0040	0.0042	0.0048
	(1.18)	(1.04)	(1.05)
$Stocks\_Traded_{q,t}$	-0.2532***	-0.2455***	-0.2347***
	(-6.95)	(-6.74)	(-6.66)
Constant	67.2828*	62.9027	53.2689
	(2.19)	(1.74)	(1.35)
Observations	180	180	180
R-squared	0.3800	0.3551	0.3354

T-statistics in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 12

Regressions of portfolio diversification potential on stock market concentration (HHI) at t-2, t-1 and t

This table reports the relation between portfolio diversification potential and stock market concentration by regressing them on the macroeconomic control variables. The sample lasts for a 45-year period from 1975 to 2019 on a quarterly basis. The dependent variables are the inverse of normalized portfolio variance of value-weighted market portfolio. Independent variables in this figure are lagged Herfindahl-Hirschman Index. All the variables are on the quarterly frequency. The t-statistics in parentheses are based on robust standard errors. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)
$HHI_{q,t-2}$	-3092.7130**		
	(-2.77)		
$HHI_{q,t-1}$		-3267.5920**	
		(-3.21)	
$HHI_{q,t}$			-3633.2080***
			(-3.36)
$Inflation_{q,t}$	-3.5957	-3.9560	-3.2109
	(-0.90)	(-0.96)	(-0.77)
$GDP\_Growth_{q,t}$	-0.0521	-0.1127	-0.1533
	(-0.07)	(-0.15)	(-0.21)
$Listed\_Firms_{q,t}$	0.0074*	0.0077*	0.0083**
	(2.11)	(2.45)	(2.82)
$Stocks\_Traded_{q,t}$	-0.2226***	-0.2200***	-0.2172***
	(-6.84)	(-6.84)	(-6.93)
Constant	29.1396	27.4332	23.4506
	(1.01)	(1.06)	(0.96)
Observations	180	180	180
R-squared	0.3245	0.3290	0.3369

T-statistics in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 13

Regressions of portfolio diversification potential (based on equal-weighted market portfolio) on stock market concentration at t-5, t-4 and t-3

This table reports the relation between portfolio diversification potential and stock market concentration by regressing them on the macroeconomic control variables. The sample lasts for a 45-year period from 1975 to 2019 on a quarterly basis. The dependent variables are the inverse of normalized portfolio variance of equal-weighted market portfolio. All the variables are on the quarterly frequency. The t-statistics in parentheses are based on robust standard errors. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
$Mkt\_Con\_Top5_{q,t-5}$	-659.9422**					
	(-3.32)					
$Mkt\_Con\_Top10_{q,t-5}$		-458.4866**				
		(-2.85)				
$Mkt\_Con\_Top5_{q,t-4}$			-657.2130**			
			(-2.84)			
$Mkt\_Con\_Top10_{q,t-4}$				-427.9951*		
				(-2.13)		
$Mkt\_Con\_Top5_{q,t-3}$					-657.7944**	
					(-2.73)	
$Mkt\_Con\_Top10_{q,t-3}$						-389.7167*
						(-1.99)
$Inflation_{q,t}$	-0.5669	-1.9884	-2.0221	-5.4817	-2.8641	-8.8367
	(-0.08)	(-0.26)	(-0.28)	(-0.74)	(-0.38)	(-1.24)
$GDP\_Growth_{q,t}$	0.5317	0.7542	0.4149	0.8444	1.1463	1.0066
	(0.41)	(0.52)	(0.29)	(0.49)	(0.67)	(0.52)
$Listed\_Firms_{q,t}$	0.0162*	0.0172*	0.0153	0.0167	0.0142	0.0169
	(2.24)	(2.24)	(1.79)	(1.70)	(1.54)	(1.60)
$Stocks\_Traded_{q,t}$	-0.5707***	-0.5308***	-0.2310***	-0.5262***	-0.5519***	-0.5215***
	(-7.92)	(-8.26)	(-8.14)	(-8.48)	(-8.26)	(-8.62)
Constant	94.2882	85.9561	99.2948	84.5477	102.8521	77.8702
	(1.42)	(1.19)	(1.26)	(0.91)	(1.23)	(0.8)
Observations	180	180	180	180	180	180
R-squared	0.4350	0.4289	0.4305	0.4226	0.4272	0.4180

T-statistics in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table 14

Regressions of portfolio diversification potential (based on equal-weighted market portfolio) on stock market concentration at t-2, t-1 and t

This table reports the relation between portfolio diversification potential and stock market concentration by regressing them on the macroeconomic control variables. The sample lasts for a 45-year period from 1975 to 2019 on a quarterly basis. The dependent variables are the normalized portfolio variance of equal-weighted market portfolio. All the variables are on the quarterly frequency. The t-statistics in parentheses are based on robust standard errors. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
$Mkt\_Con\_Top5_{q,t-2}$	-547.2294**					
	(-2.74)					
$Mkt\_Con\_Top10_{q,t-2}$		-353.8231**				
		(-2.68)				
$Mkt\_Con\_Top5_{q,t-1}$			-655.4744***			
			(-3.61)			
$Mkt\_Con\_Top10_{q,t-1}$				-428.8833***		
				(-3.35)		
$Mkt\_Con\_Top5_{q,t}$					-557.2674***	
					(-3.42)	
$Mkt\_Con\_Top10_{q,t}$						-425.8457***
						(-3.53)
$Inflation_{q,t}$	-1.6110	-6.2465	-0.4564	-3.2215	-2.4694	-0.7644
	(-0.20)	(-0.78)	(-0.05)	(-0.38)	(-0.29)	(-0.09)
$GDP\_Growth_{q,t}$	0.6223	0.3228	0.9855	0.1418	0.3041	-0.4592
	(0.39)	(0.21)	(0.66)	(0.10)	(0.20)	(-0.31)
$Listed\_Firms_{q,t}$	0.0181*	0.0197*	0.0171*	0.0198**	0.0212**	0.0232***
	(2.25)	(2.56)	(2.50)	(3.07)	(3.24)	(3.65)
$Stocks\_Traded_{q,t}$	-0.5291***	-0.5115***	-0.5295***	-0.5097***	-0.5174***	-0.5018***
	(-8.47)	(-8.58)	(-8.42)	(-8.50)	(-8.46)	(-8.58)
Constant	60.4064	50.5545	75.1080	57.9053	38.0798	32.0934
	(0.88)	(0.77)	(1.28)	(1.06)	(0.72)	(0.63)
Observations	180	180	180	180	180	180
R-squared	0.4194	0.4163	0.4340	0.4256	0.4303	0.4294

T-statistics in parentheses

*** p<0.01, ** p<0.05, * p<0.1

9 Codes

```
****** Data Processing
import pandas as pd
import numpy as np
import scipy
import datetime
from scipy import stats
import statsmodels.api as sm
import matplotlib.pyplot as plt

# Import the data set
crsp = pd.read_csv("/Users/pro/Documents/Thesis/Data/CRSP-Stocks-1970.csv")
# Drop missing values
crsp=crsp.dropna()
# Generate Variable of Market Capitalization
crsp['mkt_cap'] = crsp['PRC'] * crsp['SHROUT']
#Transform date(series) to datetime
from datetime import datetime
crsp['Date']= pd.to_datetime(crsp['date'], format = '%Y%m%d')
crsp['Year'] = crsp['Date'].dt.year
crsp['Month']= crsp['Date'].dt.month
crsp['Day']= crsp['Date'].dt.day

# Get the data for December (Yearly)
Dec_data = crsp[crsp.Date.dt.month == 12]
# Get the market capitalization for each firm at each year end
mkt_cap = Dec_data.sort_values('Date').groupby(['TICKER','Year']).tail(1)
# Get Quarterly Data
Quarterly = crsp[crsp.Date.dt.month.isin([3,6,9,12]) ]
# Get mc for each firm at each month end for each year
mkt_cap_Quar =
Quarterly.sort_values('Date').groupby(['TICKER','Year','Month']).tail(1)

# Get Number of listed firms per quarter
Num_listed = mkt_cap_Quar.groupby(['Year','Quarter'])['TICKER'].apply(lambda
grp: grp.count())
Num_listed = Num_listed.reset_index()

# Drop columns for Quarterly data
mkt_cap_Quar.drop('PRC', axis = 1, inplace = True)
```

```
mkt_cap_Quar.drop('RET', axis = 1, inplace = True)
mkt_cap_Quar.drop('SHROUT', axis = 1, inplace = True)

## Get Quarterly Market Concentration Ratios
# Total market capitalization for each quarter of each year
mkt_cap_Quar['total_mc'] =
mkt_cap_Quar.groupby(['Year','Month'])['mkt_cap'].transform('sum')
# The proportion of each firm's mc on total mc
mkt_cap_Quar['percentage_mc'] =
(mkt_cap_Quar['mkt_cap']/mkt_cap_Quar['total_mc'])

# Top 5 (Quarterly)
top5_Q = mkt_cap_Quar.groupby(['Year','Quarter'])['percentage_mc'].apply(lambda
grp: grp.nlargest(5).sum())
top5_Q = top5_Q.reset_index()
# Top 10 (Quarterly)
top10_Q = mkt_cap_Quar.groupby(['Year','Quarter'])['percentage_mc'].apply(lambda
grp: grp.nlargest(10).sum())
top10_Q = top10_Q.reset_index()

#### Reorganization of Top 5 & Top 10 (Quarterly)
# Lagged Values (5 years)
top5_Q_L5 = top5_Q.reset_index()
top5_Q_L5['MC_top5_L5'] = top5_Q['percentage_mc'].shift(20)
top10_Q_L5 = top10_Q.reset_index()
top10_Q_L5['MC_top10_L5'] = top10_Q['percentage_mc'].shift(20)

# Lagged Values (4 years)
top5_Q_L4 = top5_Q.reset_index()
top5_Q_L4 = top5_Q.drop(top5_Q[top5_Q.Year < 1971].index)
top5_Q_L4['MC_top5_L4'] = top5_Q_L4['percentage_mc'].shift(16)
top10_Q_L4 = top10_Q.drop(top10_Q[top10_Q.Year < 1971].index)
top10_Q_L4 = top10_Q_L4.reset_index()
top10_Q_L4['MC_top10_L4'] = top10_Q_L4['percentage_mc'].shift(16)

# Lagged Values (3 years)
top5_Q_L3 = top5_Q.reset_index()
top5_Q_L3 = top5_Q.drop(top5_Q[top5_Q.Year < 1972].index)
top5_Q_L3['MC_top5_L3'] = top5_Q_L3['percentage_mc'].shift(12)
top10_Q_L3 = top10_Q.drop(top10_Q[top10_Q.Year < 1972].index)
top10_Q_L3 = top10_Q_L3.reset_index()
top10_Q_L3['MC_top10_L3'] = top10_Q_L3['percentage_mc'].shift(12)
```

```
# Lagged Values (2 years)
top5_Q_L2 = top5_Q.reset_index()
top5_Q_L2 = top5_Q.drop(top5_Q[top5_Q.Year < 1973].index)
top5_Q_L2['MC_top5_L2'] = top5_Q_L2['percentage_mc'].shift(8)
top10_Q_L2 = top10_Q.drop(top10_Q[top10_Q.Year < 1973].index)
top10_Q_L2 = top10_Q_L2.reset_index()
top10_Q_L2['MC_top10_L2'] = top10_Q_L2['percentage_mc'].shift(8)

# Lagged Values (1 years)
top5_Q_L1 = top5_Q.reset_index()
top5_Q_L1 = top5_Q.drop(top5_Q[top5_Q.Year < 1974].index)
top5_Q_L1['MC_top5_L1'] = top5_Q_L1['percentage_mc'].shift(4)
top10_Q_L1 = top10_Q.drop(top10_Q[top10_Q.Year < 1974].index)
top10_Q_L1 = top10_Q_L1.reset_index()
top10_Q_L1['MC_top10_L1'] = top10_Q_L1['percentage_mc'].shift(4)

# HHI
HHI_Q_L5 = HHI_Q.reset_index()
HHI_Q_L5['HHI_L5'] = HHI_Q_L5['srq_mc'].shift(20)

HHI_Q_L4 = HHI_Q.reset_index()
HHI_Q_L4 = HHI_Q.drop(HHI_Q[HHI_Q.Year < 1971].index)
HHI_Q_L4['HHI_L4'] = HHI_Q_L4['srq_mc'].shift(16)

HHI_Q_L3 = HHI_Q.reset_index()
HHI_Q_L3 = HHI_Q.drop(HHI_Q[HHI_Q.Year < 1972].index)
HHI_Q_L3['HHI_L3'] = HHI_Q_L3['srq_mc'].shift(12)

HHI_Q_L2 = HHI_Q.reset_index()
HHI_Q_L2 = HHI_Q.drop(HHI_Q[HHI_Q.Year < 1973].index)
HHI_Q_L2['HHI_L2'] = HHI_Q_L2['srq_mc'].shift(8)

HHI_Q_L1 = HHI_Q.reset_index()
HHI_Q_L1 = HHI_Q.drop(HHI_Q[HHI_Q.Year < 1974].index)
HHI_Q_L1['HHI_L1'] = HHI_Q_L1['srq_mc'].shift(4)


# Identify the Quarter
crsp['Quarter']= crsp['Date'].dt.quarter

#Calculate the stock variance (Quarterly)
```

```
import statistics
crsp.drop(crsp[crsp['RET'] == 'C'].index, inplace = True)
crsp['RET'] = crsp['RET'].astype(float)
crsp['Quarterly_variance'] =
crsp.groupby(['Year','Quarter','TICKER'])['RET'].transform('var')
# Calculate the average per year of stock variance
ave_var_Q = crsp.groupby(['Year','Quarter'])['Quarterly_variance'].apply(lambda grp:
grp.mean())
ave_var_Q = ave_var_Q.reset_index()
# Rename
ave_var_Q = ave_var_Q.rename(columns={'Quarterly_variance':'ave_var_Q'})


market = pd.read_csv("MarketIndex_74.csv")
# Obtain the year and the month
from datetime import datetime
market['DATE']= pd.to_datetime(market['DATE'], format = '%Y%m%d')
market['Year'] = market['DATE'].dt.year
market['Month']= market['DATE'].dt.month
market['Day']= market['DATE'].dt.day
market['Quarter'] = market['DATE'].dt.quarter

market['Q'] = pd.PeriodIndex(market['DATE'], freq='Q')
#### Quarterly
# Market portfolio variance (Value-weighted)
value_var_Q = market.groupby(['Year','Quarter'])['vwretd'].apply(lambda grp:
grp.var())
value_var_Q = value_var_Q.reset_index()

# Market portfolio variance (Equal-weighted)
equal_var_Q = market.groupby(['Year','Quarter'])['ewretd'].apply(lambda grp:
grp.var())
equal_var_Q = equal_var_Q.reset_index()

# Rename
value_var_Q = value_var_Q.rename(columns={'vwretd':'value_var_Q'})
equal_var_Q = equal_var_Q.rename(columns={'ewretd':'equal_var_Q'})


# Merge Average variance & Market portfolio variance (QUARTERLY)
# With Value-weighted data
result_Q = pd.merge(ave_var_Q, value_var_Q, on=['Year','Quarter'])
```

```
# With Equal-weighted data
result_Q = pd.merge(result_Q, equal_var_Q, on=['Year','Quarter'])

# Construct Portfolio Diversification (Quarterly)
result_Q['Port_Div_val'] = result_Q['value_var_Q']/result_Q['ave_var_Q']
result_Q['Port_Div_equ'] = result_Q['equal_var_Q']/result_Q['ave_var_Q']

#### (Quarterly) Merge into Final Results (Y alreay included)
# Top 5 & Top 10 (X)
### Lagged 5 Years
result_Q = pd.merge(result_Q, top5_Q_L5, on=['Year','Quarter'])
result_Q = pd.merge(result_Q, top10_Q_L5, on=['Year','Quarter'])
### Lagged 4 Years
result_Q = pd.merge(result_Q, top5_Q_L4, on=['Year','Quarter'])
result_Q = pd.merge(result_Q, top10_Q_L4, on=['Year','Quarter'])
### Lagged 3 Years
result_Q = pd.merge(result_Q, top5_Q_L3, on=['Year','Quarter'])
result_Q = pd.merge(result_Q, top10_Q_L3, on=['Year','Quarter'])
### Lagged 2 Years
result_Q = pd.merge(result_Q, top5_Q_L2, on=['Year','Quarter'])
result_Q = pd.merge(result_Q, top10_Q_L2, on=['Year','Quarter'])
### Lagged 1 Year
result_Q = pd.merge(result_Q, top5_Q_L1, on=['Year','Quarter'])
result_Q = pd.merge(result_Q, top10_Q_L1, on=['Year','Quarter'])
```

```
***** Plotting & Regressions (Stata)
//////////////////////////////////////////////////////

//Regressions
******* Lagged 5 Years ********
**** Portfolio Diversification with Value-weighted market portfolio
** Top 5
reg d_val mc_top5_l5 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l1
outreg2 using myreg, word replace
** Top 10
reg d_val mc_top10_l5 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l2
```

```
outreg2 using myreg, word replace

**** Portfolio Diversification with Equal-weighted market portfolio
** Top 5
reg d_equ mc_top5_l5 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l3
outreg2 using myreg, word replace
** Top 10
reg d_equ mc_top10_l5 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l4
outreg2 using myreg, word replace
esttab l1 l2 l3 l4, b(%9.4f) se scalars(N r2 p)mtitles title("Lagged 5 years")
outreg2 using myreg, word dec(3)


******* Lagged 4 Years ********
**** Portfolio Diversification with Value-weighted market portfolio
** Top 5
reg d_val mc_top5_l4 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l1
outreg2 using myreg, word replace
** Top 10
reg d_val mc_top10_l4 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l2
outreg2 using myreg, word replace


**** Portfolio Diversification with Equal-weighted market portfolio
** Top 5
reg d_equ mc_top5_l4 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l3
outreg2 using myreg, word replace
** Top 10
reg d_equ mc_top10_l4 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l4
outreg2 using myreg, word replace
esttab l1 l2 l3 l4, b(%9.4f) se scalars(N r2 p)mtitles title("Lagged 4 years")
outreg2 using myreg, word dec(3)


******* Lagged 3 Years ********
**** Portfolio Diversification with Value-weighted market portfolio
```

```
** Top 5
reg d_val mc_top5_l3 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l1
outreg2 using myreg, word replace
** Top 10
reg d_val mc_top10_l3 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l2
outreg2 using myreg, word replace


**** Portfolio Diversification with Equal-weighted market portfolio
** Top 5
reg d_equ mc_top5_l3 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l3
outreg2 using myreg, word replace
** Top 10
reg d_equ mc_top10_l3 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l4
outreg2 using myreg, word replace
esttab l1 l2 l3 l4, b(%9.4f) se scalars(N r2 p)mtitles title("Lagged 3 years")
outreg2 using myreg, word dec(3)

******* Lagged 2 Years ********
**** Portfolio Diversification with Value-weighted market portfolio
** Top 5
reg d_val mc_top5_l2 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l1
outreg2 using myreg, word replace
** Top 10
reg d_val mc_top10_l2 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l2
outreg2 using myreg, word replace


**** Portfolio Diversification with Equal-weighted market portfolio
** Top 5
reg d_equ mc_top5_l2 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l3
outreg2 using myreg, word replace
** Top 10
reg d_equ mc_top10_l2 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l4
```

```
outreg2 using myreg, word replace
esttab l1 l2 l3 l4, b(%9.4f) se scalars(N r2 p)mtitles title("Lagged 2 years")
outreg2 using myreg, word dec(3)


******* Lagged 1 Year1 ********
**** Portfolio Diversification with Value-weighted market portfolio
** Top 5
reg d_val mc_top5_l1 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l1
outreg2 using myreg, word replace
** Top 10
reg d_val mc_top10_l1 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l2
outreg2 using myreg, word replace


**** Portfolio Diversification with Equal-weighted market portfolio
** Top 5
reg d_equ mc_top5_l1 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l3
outreg2 using myreg, word replace
** Top 10
reg d_equ mc_top10_l1 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l4
outreg2 using myreg, word replace
esttab l1 l2 l3 l4, b(%9.4f) se scalars(N r2 p)mtitles title("Lagged 1 year")
outreg2 using myreg, word dec(3)



///// No lag///////
**** Portfolio Diversification with Value-weighted market portfolio
** Top 5
reg d_val mc_top5 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l1
outreg2 using myreg, word replace
** Top 10
reg d_val mc_top10 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l2
outreg2 using myreg, word replace
```

```
**** Portfolio Diversification with Equal-weighted market portfolio
** Top 5
reg d_equ mc_top5 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l3
outreg2 using myreg, word replace
** Top 10
reg d_equ mc_top10 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l4
outreg2 using myreg, word replace
esttab l1 l2 l3 l4, b(%9.4f) se scalars(N r2 p)mtitles title("Lagged 0 years")
outreg2 using myreg, word dec(3)

////////////////////////////////////////////////////

//HHI
** lag 5
reg d_val hhi_l5 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l1
outreg2 using myreg, word replace

reg d_equ hhi_l5 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l2
outreg2 using myreg, word replace

** lag 4
reg d_val hhi_l4 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l3
outreg2 using myreg, word replace

reg d_equ hhi_l4 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l4
outreg2 using myreg, word replace

** lag 3
reg d_val hhi_l3 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l5
outreg2 using myreg, word replace

reg d_equ hhi_l3 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l6
outreg2 using myreg, word replace
```

```
esttab l1 l2 l3 l4 l5 l6, b(%9.4f) se scalars(N r2 p)mtitles title("HHI 5-3")
outreg2 using myreg, word dec(3)

**lag 2
reg d_val hhi_l2 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l1
outreg2 using myreg, word replace

reg d_equ hhi_l2 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l2
outreg2 using myreg, word replace

** lag 1
reg d_val hhi_l1 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l3
outreg2 using myreg, word replace

reg d_equ hhi_l1 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l4
outreg2 using myreg, word replace

** NO lag
reg d_val hhi inflation_1 gdpgrowth num_listed stockstraded, robust
est store l5
outreg2 using myreg, word replace

reg d_equ hhi inflation_1 gdpgrowth num_listed stockstraded, robust
est store l6
outreg2 using myreg, word replace

esttab l1 l2 l3 l4 l5 l6, b(%9.4f) se scalars(N r2 p)mtitles title("HHI 2-0")
outreg2 using myreg, word dec(3)
//summarize

//////////////////////////////////////////////////////
// Average variance of individual stocks
******* Lag 5
** Top 5
reg ave_var_q mc_top5_l5 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l1
outreg2 using myreg, word replace
```

```
** Top 10
reg ave_var_q mc_top10_l5 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l2

******* Lag 4
** Top 5
reg ave_var_q mc_top5_l4 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l3
outreg2 using myreg, word replace
** Top 10
reg ave_var_q mc_top10_l4 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l4
outreg2 using myreg, word replace

******* Lag 3
** Top 5
reg ave_var_q mc_top5_l3 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l5
outreg2 using myreg, word replace
** Top 10
reg ave_var_q mc_top10_l3 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l6
outreg2 using myreg, word replace


esttab l1 l2 l3 l4 l5 l6, b(%9.4f) se scalars(N r2 p)mtitles title("ave_var_q 5-3")
outreg2 using myreg, word dec(3)


******* Lag 2
** Top 5
reg ave_var_q mc_top5_l2 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l1
outreg2 using myreg, word replace
** Top 10
reg ave_var_q mc_top10_l2 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l2

******* Lag 1
** Top 5
reg ave_var_q mc_top5_l1 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l3
```

```
outreg2 using myreg, word replace
** Top 10
reg ave_var_q mc_top10_l1 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l4
outreg2 using myreg, word replace

******* No lags
** Top 5
reg ave_var_q mc_top5 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l5
outreg2 using myreg, word replace
** Top 10
reg ave_var_q mc_top10 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l6
outreg2 using myreg, word replace


esttab l1 l2 l3 l4 l5 l6, b(%9.4f) se scalars(N r2 p)mtitles title("ave_var_q 5-3")
outreg2 using myreg, word dec(3)

/////////////////////////////////////////////////////

///// Market Portfolio Variance (Value)
******* Lag 5
** Top 5
reg value_var_q mc_top5_l5 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l1
outreg2 using myreg, word replace
** Top 10
reg value_var_q mc_top10_l5 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l2

******* Lag 4
** Top 5
reg value_var_q mc_top5_l4 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l3
outreg2 using myreg, word replace
** Top 10
reg value_var_q mc_top10_l4 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l4
outreg2 using myreg, word replace
```

```
******* Lag 3
** Top 5
reg value_var_q mc_top5_l3 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l5
outreg2 using myreg, word replace
** Top 10
reg value_var_q mc_top10_l3 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l6
outreg2 using myreg, word replace


esttab l1 l2 l3 l4 l5 l6, b(%9.4f) se scalars(N r2 p)mtitles title("value_var_q 5-3")
outreg2 using myreg, word dec(3)

******* Lag 2
** Top 5
reg value_var_q mc_top5_l2 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l1
outreg2 using myreg, word replace
** Top 10
reg value_var_q mc_top10_l2 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l2

******* Lag 1
** Top 5
reg value_var_q mc_top5_l1 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l3
outreg2 using myreg, word replace
** Top 10
reg value_var_q mc_top10_l1 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l4
outreg2 using myreg, word replace

******* No lags
** Top 5
reg value_var_q mc_top5 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l5
outreg2 using myreg, word replace
** Top 10
reg value_var_q mc_top10 inflation_1 gdpgrowth num_listed stockstraded, robust
est store l6
outreg2 using myreg, word replace
```

```
esttab l1 l2 l3 l4 l5 l6, b(%9.4f) se scalars(N r2 p)mtitles title("value_var_q 2-0")
outreg2 using myreg, word dec(3)

//////////////////////////////////////////////////////
////////Average volatility
//Test for Structural Breaks
**Generate Quarterly Time Variable
gen t=tq(1975q1)+_n-1
format t %tq

** Find the structural Break
reg ave_var_q t
tsset t
estat sbsingle

** Plot
gen upper1 = 0.008
twoway bar upper1 t if inrange(t, 80, 82)|inrange(t, 86, 91)|inrange(t, 122,
124)|inrange(t, 164, 167)|inrange(t, 191, 197), bcolor(gs12) || tsline ave_var_q,
tline(2003q2) xtitle("Time") ytitle("Average Variance of Individual Stocks")
title("Average Variance of Individual Stocks over Time") lcolor(navy) legend(off)
tlabel(, format(%tq))

** Trend Regression
gen pre = 0 if t>q(2003q2)
replace pre = 1 if t<q(2003q2)
reg ave_var_q t ave_var_l1 if pre == 1, robust
reg ave_var_q t ave_var_l1 if pre == 0, robust




// Market Volatility - Value-weighted
** Structural Breaks
reg value_var_q t
tsset t
estat sbsingle

** Plotting
tsline value_var_q
```

```
gen upper = 0.002
twoway bar upper t if inrange(t, 80, 82)|inrange(t, 86, 91)|inrange(t, 122,
124)|inrange(t, 164, 167)|inrange(t, 191, 197), bcolor(gs12) || tsline value_var_q,
tline(2008q3) xtitle("Time") ytitle("Market Portfolio Variance") title("Market
Portfolio Variance over Time") lcolor(navy) legend(off) tlabel(, format(%tq))
** Trend Regression
gen mkt_pre = 0 if t>q(2008q3)
replace mkt_pre = 1 if t<q(2008q3)
reg value_var_q t value_var_l1 if mkt_pre == 1, robust
reg value_var_q t value_var_l1 if mkt_pre == 0, robust


tsline num_listed, xtitle("Time") ytitle("Number of listed firms") title("Number of
listed firms in stock market over Time")
```